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Portfolio

Selected research posters and
project reports

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Action Identifiability in Protein-Design Agents 2
This MoML 2026 poster with Gabriela Lobinska shows that ESM-2 compresses distinct mutational processes by 120 times and that a protein-design agent cannot reliably identify the correct action.

What Can We Learn From Evolution? 3
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Single-Cell RNA Sequencing in Developing Human Brains 5
This poster from the Pattwell Lab at Seattle Children's Research Institute won the People's Choice Award at the 2024 Stanford Comprehensive Cancer Research Training Program and applies machine learning to developing-brain data to investigate the origins of medulloblastoma.

Harvesting Power at High Frequencies 6
This poster won 1st Place, the AFCEA Excellence in Technology award, and the Sigma Xi Best Interdisciplinary Research award at the 2024 Washington State Science and Engineering Fair, and it tests an RF power combiner and rectarray as an alternative to the traditional battery.

Lights, Camera, Action! 7
This history poster with Maxwell Yu uses archival photographs and documents to show how Los Angeles and Hollywood were illuminated in the early 1900s.

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This DECA Community Giving Project with Prisha Kuberan and Kaustubh Grama raised over $22,000 for cataract surgeries and won 1st Place at the International Career Development Conference.

Integrated Marketing Campaign for SWENext 30
This 10-page DECA campaign plan with Kaustubh Grama promotes SWENext, the Society of Women Engineers K-12 program, during Women's History Month.

Action Identifiability in Protein-Design Agents

Saanvi S. Subramanian[1*], Gabriela Lobinska[2]
[1]California Institute of Technology, [2]AITHYRA
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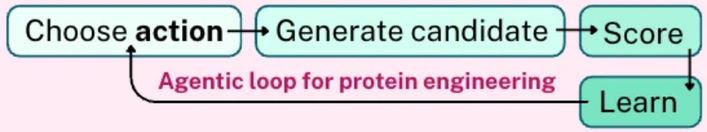
MoML @ MIT 2026

How can we ensure agents make informed, well-reasoned decisions in protein design?



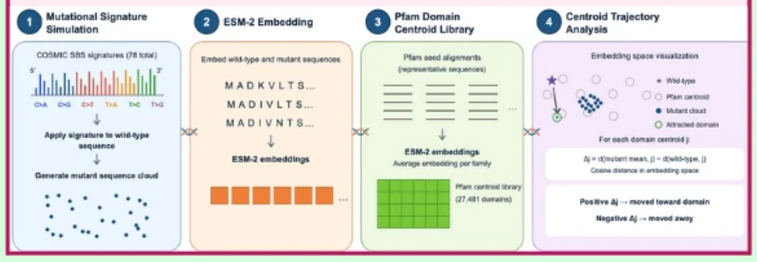
Does the scorer preserve enough information to distinguish the available actions?

Action Identifiability:

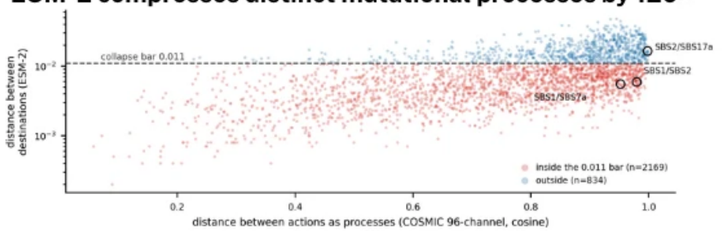
The ability to distinguish the outcomes of competing actions using the agent's scoring model.

Before selecting molecular interventions, agents should establish action identifiability by ***verifying that their scoring models can distinguish available actions.***

We built a controlled protein-design environment, using COSMIC mutational signatures as actions, to test this:



ESM-2 compresses distinct mutational processes by 120×

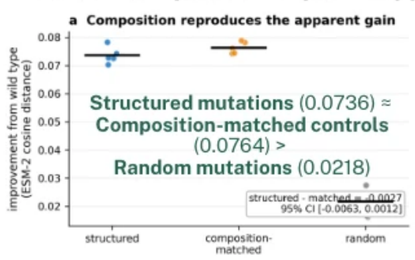


Compared 3,003 pairs of COSMIC mutational signatures before and after applying them to proteins and embedding the outcomes with ESM-2

Median pairwise distance:
COSMIC signature space 0.815
ESM-2 outcome space 0.0068

59.3% of action pairs are widely separated in COSMIC space (>0.5) but nearly overlap in ESM-2 space (<0.011)

Mutation composition explains apparent optimization gains



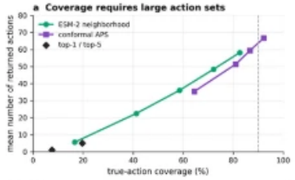
Does the model look at specific mutational process or just which substitutions occur?

Control: Preserve substitutions but randomize their positions and sequence contexts

Reproduces the optimization gain without preserving the original mutation pattern.

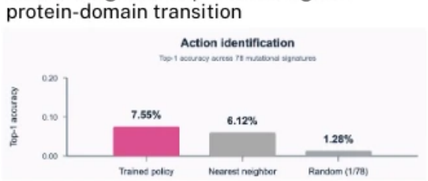
Agents cannot reliably identify the correct action

Can uncertainty quantification recover reliable decisions?



92.2% coverage → 66.8/78 actions returned

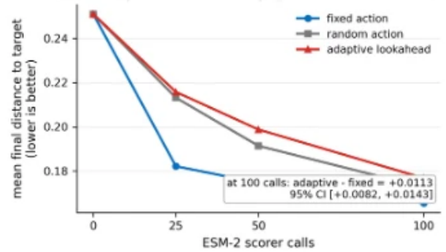
Trained a policy to predict which of the 78 COSMIC signatures produced a given protein-domain transition



7.55% accuracy → 92.45% incorrect

The agent cannot reliably identify a specific action (mutational process), even when allowed to return multiple candidate actions.

Adaptive action selection does not improve optimization



Compared adaptive five-action lookahead against fixed and random action selection under equal ESM-2 scoring budgets.

Policy	Distance
Fixed action	0.165
Random actions	0.173
Adaptive lookahead	0.177

Evaluating more candidate actions does not improve optimization when the feedback provides insufficient useful information.

Improving ESM-2 scores does not establish target-domain acquisition

Does moving closer to a target in ESM-2 space produce proteins with the target domain? → **0** evaluated sequences passed Pfam's curated threshold for **PF17041**.

Key takeaways

- Distinct actions become indistinguishable
- Agentic decision-making does not overcome poor action resolution
- Optimization requires independent validation

Scan to view project page for paper, code & results!

Action Identifiability in Protein-Design Agents

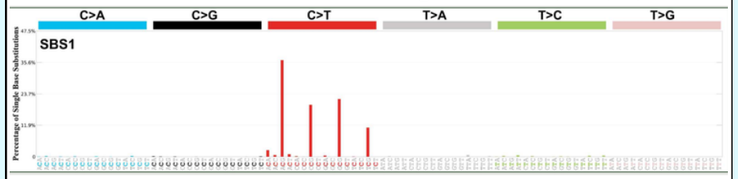
This MoML 2026 poster with Gabriela Lobinska shows that ESM-2 compresses distinct mutational processes by 120 times and that a protein-design agent cannot reliably identify the correct action.

What Can We Learn From Evolution? Simulating Functional Effects of Mutation Signatures on Protein Sequence Generation with ESM-2

Saanvi Skanda Subramanian[1]*, Gabriela Lobinska[2]
[1]California Institute of Technology, [2]AITHYRA

PROJECT DESCRIPTION

- COSMIC = Catalogue of Somatic Mutations in Cancer
  - Database of different mutational signatures (SBS, DBS, ID)



- ***Do mutation signatures push proteins toward specific alternative functions?*** **PRISM aims to test this directional pressure**
  - **ESM-2** maps WT, mutant proteins into a shared 320-D space
  - **Pfam** centroids provide reference points for 27,481 protein fams

DATA & MODELS

78 COSMIC SBS signatures	**3** Pfam structural clans	**100** mutants/ signature/protein	**8M** ESM-2 parameters

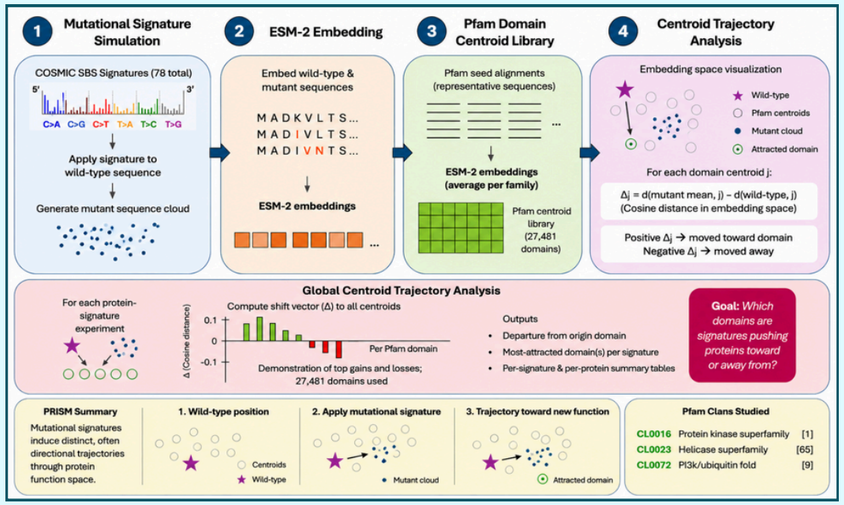
- **Clans:** kinases, helicases, PI3K/ubiquitin
- **Protein pairs share fold, EC class structure, and >60% sequence identity**

SCORING CENTROID ATTRACTION

$$\Delta j = d(WT, j) - d(mutant\ cloud, j)$$

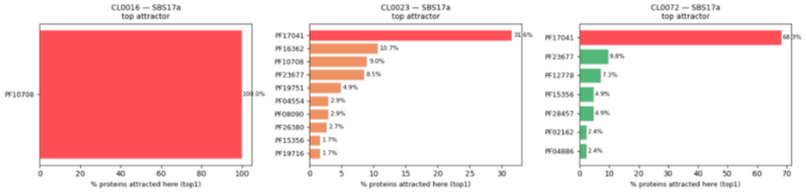
positive Δj → attraction toward family j

Rank the nearest Pfam family for every mutant cloud, repeat across independent clans and null permutations



MAIN RESULT: SBS17a → PF17041

SBS17a consistently attracts sequences toward PF17041 (SasG_E domain)



- Short bacterial surface domain adhesion and biofilm function
- Directionality demonstrates against random drift

What drives this movement? Composition explains most observed displacement

VALIDATION & CONTROLS

10,000 null permutations family-wise p<0.0001	**72%** movement retained matched scramble	**18.7×** length-adjusted enrichment from 60.7× raw
5.0 SD beyond natural-protein floor metric-optimization endpoint	**0.7%** held-out top-1 accuracy signature-grouped evaluation	

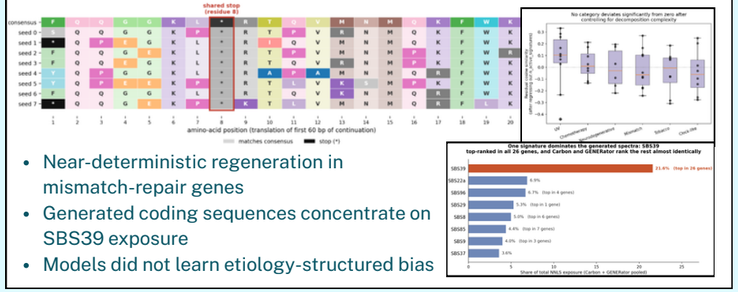
GENERATIVE-MODEL EXTENSION

Extending PRISM: Do generative models show the same bias?

26 genes	**50** samples/model	**20%** of beginning of each CDS as prompt

- Carbon-500M and GENErator-v2-prokaryote-1.2b autoregressive models

79.3%, 73.1% translation success	**73.7%, 75.0%** mean nucleotide identity	**r = 0.86** model outcomes agree across genes



- Near-deterministic regeneration in mismatch-repair genes
- Generated coding sequences concentrate on SBS39 exposure
- Models did not learn etiology-structured bias

NEXT STEPS & REFERENCES

- Validate → PF17041 with HMMER
- Scale to 53 genes & 200 samples
- Fine-tune genomic models on selected signatures
- Train Pfam-transition predictors for sequence design

References
Alexandrov et al. Nature 2020
Lin et al. Science 2023
Mistry et al. NAR 2021

What Can We Learn From Evolution?

This CBEC 2026 poster with Gabriela Lobinska simulates 78 COSMIC mutational signatures on proteins and measures where ESM-2 moves the mutated sequences relative to 27,481 Pfam protein families.

MedulloModel: Building a Novel Machine Learning Approach to Analyze Single-cell RNA Sequencing Data to Detect Medulloblastoma Tumorigenesis

Saanvi Subramanian, Kaustubh Grama

OVERVIEW

OBJECTIVE: Model medulloblastoma cell evolution using **pseudotime analysis** and **variational autoencoders (VAEs)**

- Identify **key genetic drivers** and link **developmental trajectories** to clinical outcomes

WORKFLOW: Data collection of mice data, preprocessing, trajectory inference, deep learning, 3D visualization, validation and analysis

RESULTS:

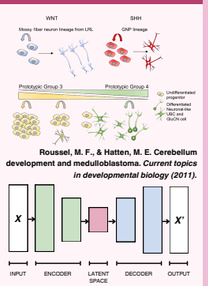
- Identified 5 distinct tumor progression trajectories.
- High-risk pathways enriched in specific genes/pathways correlated with recurrence (p-value)

CONCLUSION:

- Combining scRNA-seq, deep learning, and 3D modeling provides **new insights** into **medulloblastoma progression**
- Identified **high-risk** trajectories could help improve **prognostic predictions** and inform targeted therapies.

BACKGROUND

- **Medulloblastoma**: A highly aggressive pediatric brain tumor with distinct molecular subgroups (WNT, SHH, Group 3, Group 4)
- **Single-Cell RNA Sequencing** (scRNA-seq): Allows for analyzing individual tumor cells.
- **Pseudotime Analysis**: Reconstructs tumor cell progression from scRNA-seq data using tools like Monocle3 and scVelo to identify key genes driving evolution.
- **Variational Autoencoders (VAEs)**: A deep learning method that learns a low-dimensional representation of tumor cell states, improving trajectory inference accuracy.



Roussel, M. F., & Hatten, M. E. Cerebellum development and medulloblastoma. *Current topics in developmental biology (2011).*

Basic scheme of a Variational Autoencoder (VAE)

Graphic designed by Saanvi Subramanian using Canva

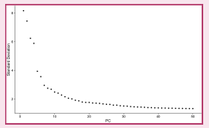
METHODOLOGY

PROJECT WORKFLOW

1. **Data Collection:** Obtain scRNA-seq data from medulloblastoma tumors
2. **Data Preprocessing:** Normalization, feature selection, dimensionality reduction
3. **Trajectory Inference:** Monocle-3 workflow, key gene identification
4. **Deep Learning-Based Modeling:** Train a Variational Autoencoder (VAE)
5. **Validation & Analysis: Perform survival analysis, evaluate clustering data**

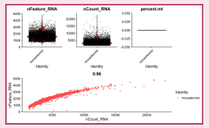
Graphic designed by Saanvi Subramanian using Canva

RESULTS



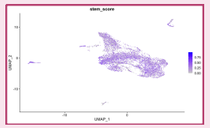
Principal Component Analysis (PCA)*:

The **elbow** in the plot around PC 10 suggests that **subsequent PCs contribute minimally** to explaining the transcriptional heterogeneity in the medulloblastoma tumor microenvironment, helping determine the optimal dimensionality for downstream analyses like clustering.



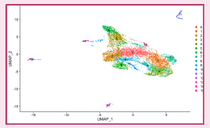
Violin and Scatter plots*:

Most cells in this mouse tumor Seurat object express **1000-2000 genes** and **2000-8000 RNA counts**. The **strong correlation (0.96)** between nCount_RNA and nFeature_RNA indicates **high-quality single-cell data** with consistent RNA capture efficiency across the tumor cells.



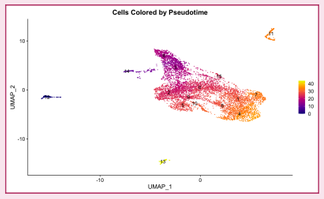
Stem Score Distribution*:

Stem_score distribution across medulloblastoma cells with **relatively moderate stemness** (purple, 0.25-0.50) throughout most cells and no clear high-stemness population. The **stem cell signature** appears **fairly uniform** across the tumor microenvironment.

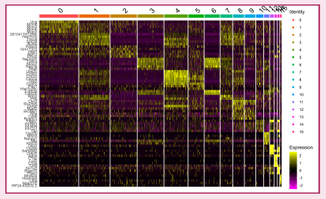


Cell Clustering*:

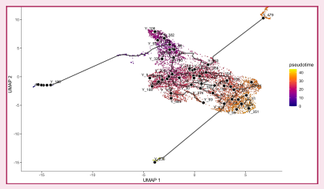
Shows **16 distinct cell clusters** in the sample, with each cluster colored differently. Major populations include **central clusters (0, 1, 2)** surrounded by various developmental trajectories, with **isolated clusters (11, 12, 13)** potentially representing rare cell types or tumor microenvironment components.



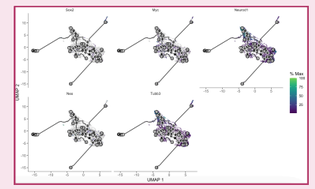
Cells Colored by Pseudotime*: Single-cell trajectory analysis showing medulloblastoma cells arranged along a **pseudotemporal progression path**, with color gradient indicating developmental stage.



Gene Expression Heatmap*: Hierarchical clustering of **differentially expressed genes** across tumor samples, revealing distinct molecular patterns associated with different **tumor progression trajectories**.



Pseudotime trajectory analysis*: Developmental progression from **early progenitors** (blue, left) to **differentiated states** (yellow/orange, right). Black connecting lines show principal differentiation paths with labeled cells.



Feature plots*: Shows expression patterns of five key neural genes (Sox2, Myc, Neurod1, Nes, Tubb3) across medulloblastoma cells. Neurod1 and Tubb3 show higher expression (purple) in differentiated regions, while stem cell markers appear more uniformly distributed.

All eight graphs produced by Kaustubh Grama using RStudio

EVALUATING PERFORMANCE

- Model achieved high accuracy in identifying tumor subtypes (WNT, SHH, Group 3, Group 4)
- **Performance metrics**: 93% classification accuracy on test set
- Trajectory inference validated using **known genetic markers**, further validation needed
- Successful identification of key genes driving progression pathways
- **Cross-validation demonstrated robustness** against batch effects
- **Survival prediction accuracy significantly higher** than conventional methods ($p<0.01$)

CONCLUSION

- MedulloModel integrates **scRNA-seq**, **pseudotime analysis**, and **deep learning** to provide insights into medulloblastoma progression
- Identified **5 distinct tumor evolution trajectories** with potential clinical relevance, though further validation in larger cohorts is necessary
- **High-risk developmental pathways** were correlated with poor prognosis and treatment resistance, supporting their role in tumor progression
- Several novel **genetic drivers identified**, which could serve as potential therapeutic targets, though experimental validation is required
- Findings offer promising insights for precision medicine approaches

FUTURE WORK

- **Expand dataset** to include more patient samples **across all subgroups**
  - Analyze human data instead of mice data
- Integrate **spatial transcriptomics data** to enhance tumor microenvironment understanding
- Develop interactive clinical decision support tool for personalized treatment planning
- Validate identified therapeutic targets through in vitro and in vivo studies
- Incorporate multi-omics data (proteomics, epigenomics) to improve model accuracy
- **Extend methodology** to other pediatric brain cancers
- Clinical trial design targeting high-risk progression pathways

REFERENCES

Cope, E. C., Briones, B. A., Brockett, A. T., Martinez, S., Vigneron, P. A., Opendak, M., Wang, S. S., & Gould, E. (2016). Immature Neurons and Radial Glia, But Not Astrocytes or Microglia, Are Altered in Adult Cntnap2 and Shank3 Mice, Models of Autism. eNeuro, 3(5), ENEURO.0196-16.2016.

Pan, J., Ma, N., Yu, B. et al. Transcriptomic profiling of microglia and astrocytes throughout aging. Neuroinflammation 17, 97 (2020).

Junyue Cao et al. A human cell atlas of fetal gene expression.Science370,eaba7721(2020).DOI:10.1126/science.aba7721

MedulloModel

This poster with Kaustubh Grama presents a machine learning approach that detects medulloblastoma tumorigenesis in single-cell RNA sequencing data.

Using Single-Cell RNA Sequencing in Developing Human Brains to Investigate the Origins of Medulloblastoma through Applying Various Machine Learning Models

Saanvi Subramanian[1,2], Ashmitha Rajendran[2,3]
[1]Edmonds College, Lynnwood, WA;[2]Seattle Children's Research Institute: Pattwell Lab;[3]Department of Bioinformatics, University of Washington

Abstract

- Early brain development helps uncover intracranial tumor mechanisms.
- RNA sequencing can identify diverse brain cell types.
- Project uses **supervised machine learning** to analyze **scRNA-seq** data from human embryogenesis and pediatric tumor datasets.
- Goal is to compare gene expression in healthy and medulloblastoma cells to uncover genetic markers and pathways involved in brain cancer development.

Introduction

- Medulloblastoma
  - Malignant brain tumor in the cerebellum
  - **20%** of all **pediatric brain tumors**
  - **63%** of intracranial embryonic tumors
  - **Cellular lineages of origin for some subtypes have not been definitively proven**
- Single-cell RNA sequencing
  - Examines gene expression from individual cells
  - Identifies cell types across tissues
  - Detects transient cell states in tumors
  - Maps cell development trajectories

About the Data

- sci-RNA-seq3 human cerebellum
- 1,092,000 cells total
- From day 89-125
- Downsampled to 2700 cells per timepoint
  - 13,500 total

Cao, J., O'Day, D. R., Pliner, H. A., Kingsley, P. D., Deng, M., Daza, R. M., Zager, M. A., Aldinger, K. A., Blecher-Gonen, R., Zhang, F., Spielmann, M., Palis, J., Doherty, D., Steemers, F. J., Glass, I. A., Trapnell, C., & Shendure, J. (2020). A human cell atlas of fetal gene expression. Science (New York, N.Y.), 370(6518), eaba7721. https://doi.org/10.1126/science.aba7721

Workflow

PHASE 1: Pre-processing data

Raw sc-RNAseq data
Normalization
Feature Selection
Scaling
Dimensionality Reduction
Processed sc-RNAseq data --> Integration

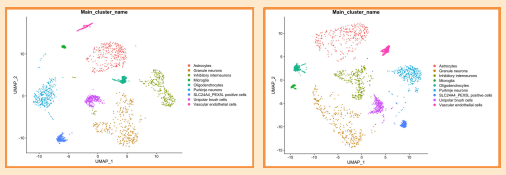
PHASE 2: Creating and testing model

Optimizing gene selection + Training model
Testing model with day 110 and 94
Evaluating model

Results

1A: Pre-processing Expression Data

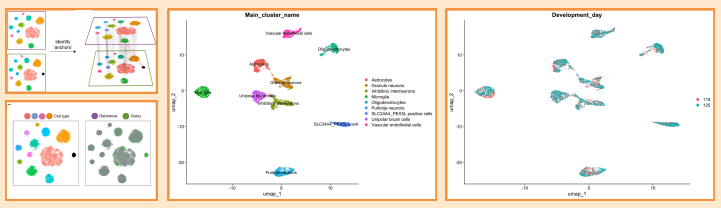
Visual analysis of developmental datasets



Uniform Manifold Approximation and Projection

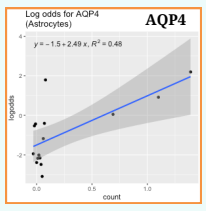
- Using day 115 and 125 data
  - most developed, most definitive cell types annotations
- Clustered by Louvain clustering, colored by cell type determined by known genetic markers

Integrating developmental datasets



1B: Identifying Top Genes for Each Cell Type

Using LogOdds to isolate genes with the largest effect on a cell being a specific cell type



*Known astrocyte markers

LogOdds plot shows if the probability a cell is an astrocyte increases directly with the increase of a gene's expression value

Using Variance Inflation Factor to examine multicollinearity

```
> car::vif(model_lm)
      PTN     DCLK2     COPE     MEIS1     GLUL
1.217501 1.083378 1.074164 1.137357 1.117989
     CST3      LMF2     DIRC3     RPL11    ANKS1B
1.155255 1.063681 1.085157 1.119860 1.128402
     XKR4      CDK6     RAD51B     RPS2      QKI
1.069174 1.082791 1.079541 1.078704 1.045264
    GPM6B       NES     ILDR2      PAX3      ENO1
1.078396 1.081792 1.084031 1.090055 1.080070
    PTPN13      LCAT      TTYH1      SATB1     CRADD
1.068951 1.084395 1.050595 1.078491 1.056830
    S100B      GFAP      APOE      AQP4
1.103309 1.055284 1.114129 1.187667
```

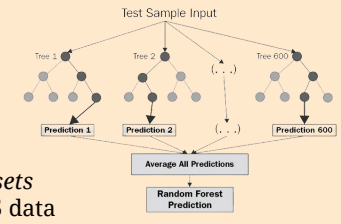
2A: Creating and Training ML Models

Logistic Regression Model

Intercept Weight Gene

$$\beta_0 + W$$

Random Forest Model

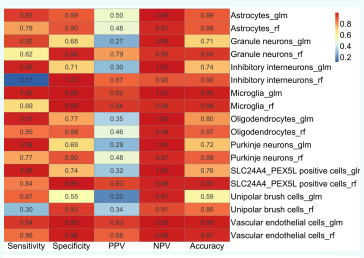


Creating Testing and Training Datasets

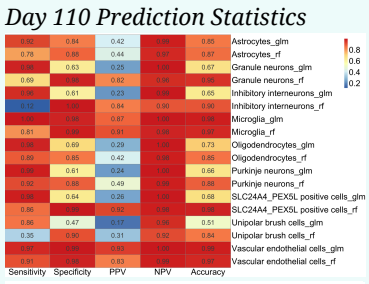
- **Trained** from 75% Day 115+125 data
- **Testing I:** 25% Day 115+125 data
- **Testing II:** Day 94, 110 data

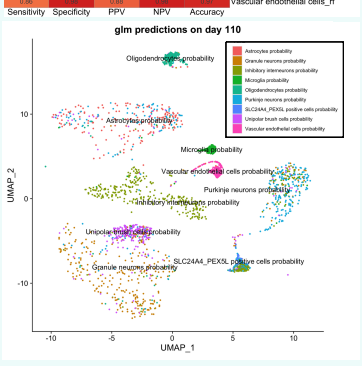
2B: Evaluating Performance

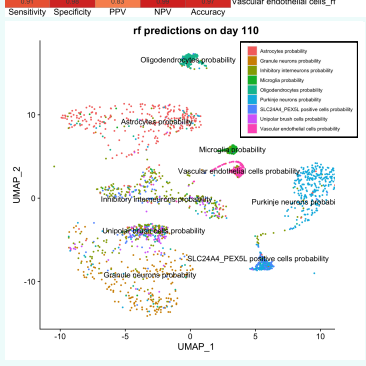
Day 94 Prediction Statistics



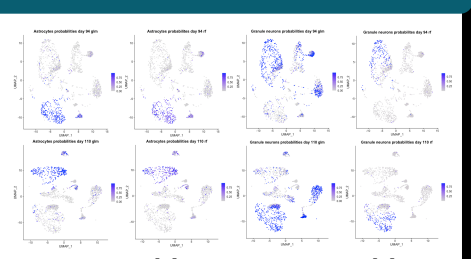
Day 110 Prediction Statistics





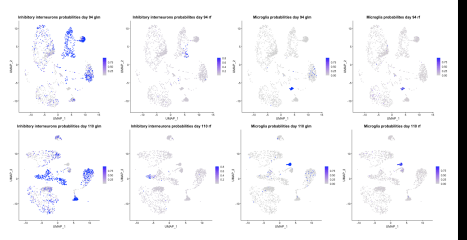


Model Analysis



GLM model predictions are more accurate to cell clusters for astrocytes than RF

RF model predictions are more accurate to cell clusters for granule neurons than GLM



Both models do not predict inhibitory interneurons well

Both models predict microglia notably well

Next Steps

- Try additional ML models and ensembling techniques
- Test models on a new cerebellum human embryonic development dataset:
  - Aldinger, K.A., Thomson, Z., Phelps, I.G. et al. Spatial and cell type transcriptional landscape of human cerebellar development. Nat Neurosci 24, 1163–1175 (2021). https://doi.org/10.1038/s41593-021-00872-y
- Apply to Medulloblastoma sequencing data

References

The GeneCards Suite: From Gene Data Mining to Disease Genome Sequence Analyses (PMID: 27322403; Citations: 3,306) Stelzer G, Rosen R, Plaschkes I, Zimmerman S, Twik M, Fishilevich S, Iny Stein T, Nudel R, Lieder I, Mazor Y, Kaplan S, Dahary, D, Warshawsky D, Guan - Golan Y, Kohn A, Rappaport N, Safran M, and Lancet D

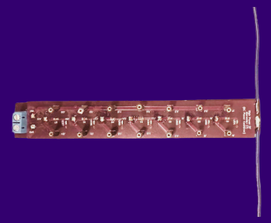
Pan, J., Ma, N., Yu, B. et al. Transcriptomic profiling of microglia and astrocytes throughout aging. J Neuroinflammation 17, 97 (2020). https://doi.org/10.1186/s12974-020-01774-9

Junyue Cao et al. A human cell atlas of fetal gene expression.Science370,eaba7721(2020).DOI:10.1126/science.aba7721

Monteagudo, A., Feola, J., Natola, H., Ji, C., Pröschel, C., & Johnson, G. V. W. (2018). Depletion of astrocytic transglutaminase 2 improves injury outcomes. Molecular and cellular neurosciences, 92, 128–136. https://doi.org/10.1016/j.mcn.2018.06.007

Single-Cell RNA Sequencing in Developing Human Brains

This poster from the Pattwell Lab at Seattle Children's Research Institute won the People's Choice Award at the 2024 Stanford Comprehensive Cancer Research Training Program and applies machine learning to developing-brain data to investigate the origins of medulloblastoma.



Harvesting Power at High Frequencies:

Building an RF Power Combiner and Rectarray as a Sustainable Alternative to the Traditional Battery

Saanvi Subramanian

INTRODUCTION

The widespread use of **traditional batteries** in modern devices comes with **significant environmental consequences**. Batteries can contain **hazardous materials** like **lead**, **cadmium**, and **mercury**, which pose **risks** to **ecosystems** and **human health,** as evidenced by the frequent emergency department visits by children due to battery-related injuries (one every 75 minutes). **Improper disposal** of these batteries leads to **soil and water pollution**, and their regular replacement contributes to **growing electronic waste**, with only 5% of lithium-ion batteries recycled in the US. Moreover, the **extraction of natural resources** required for battery production, particularly lithium, raises environmental and social concerns with resource depletion and a substantial carbon footprint.

- **NEGATIVE EFFECTS OF TRADITIONAL BATTERIES**
- **HAZARDOUS MATERIALS AND HEALTH RISKS**
- **IMPROPER DISPOSAL AND WASTE GENERATION**
- **RESOURCE EXTRACTION AND SOCIAL CONCERNS**

The purpose of this project was to create a more sustainable power supply source, after learning about the negative, harmful global impacts of traditional batteries. Energy harvesting was used to create this alternative energy source to convert ambient RF energy emitted from cell towers to a DC power output through use of seven rectifiers as a unique topology.

BACKGROUND

Energy harvesting is defined as the conversion of ambient energy present in the environment into electrical energy for use in powering autonomous electronic devices or circuits - a renewable alternative to traditional energy sources.

This research focuses on the potential of harvesting of ambient RF energy emitted from current infrastructure, to take advantage of the abundance of cellular towers and high density of carrier signals to produce power sustainably.

RESEARCH OBJECTIVES

Engineering Problem: The exploration of energy harvesting techniques within the ultra-high frequency (UHF) band represents an area that has been largely overlooked by prior research efforts. To date, no comprehensive solution has been devised to effectively harness the ambient radiofrequency (RF) energy emitted from cellular communication towers, which is continuously radiated into the environment on a daily basis.

Research Question: What is the peak power output achieved by a harvester board design with seven rectifiers when receiving signals from a UHF source?

Target Outcome: Achieve DC power output harvested from prototype in proximity to a UHF source.

Claim: This research will develop an RF harvester board utilizing seven rectifiers, along with a power combiner, to efficiently receive a simulated cell tower signal (**749 MHz**: the central carrier frequency of a local UW cell tower) and produce a DC power output. Results would be plotted and analyzed to find the peak power output harvested.

Project follows the engineering design process shown below.

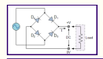
DESIGN → BUILD → TEST → ANALYZE

- **DESIGN**
  - Rectifier circuit
  - PCBs
    - *LTSpice*
    - *KiCAD*
- **BUILD**
  - Harvester board
  - Power combiner
- **TEST**
  - Use signal source
  - Run experiment
- **ANALYZE**
  - Plot results
  - Identify peak power output
  - Statistical analysis

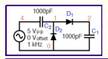
ENGINEERING METHODOLOGY

Design of Rectifier

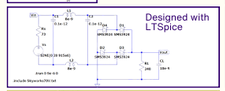
Two alternative rectifier circuit designs were evaluated using simulation tools, and the best performing design was selected for prototyping.

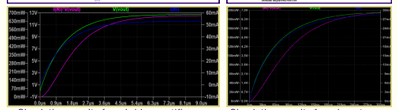


Full Wave Bridge Rectifier circuit schematic



"Booster" Rectifier circuit schematic





Simulation results from bridge rectifier | Simulation results from booster

The **full wave bridge rectifier** showed highest efficiency (>3 times as much power delivered to load), and was chosen.

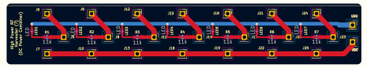
Design of Harvester Board

KiCAD Design

- Specific component details in Bill of Materials and Datasheets

Design of Power Combiner

KiCAD Design



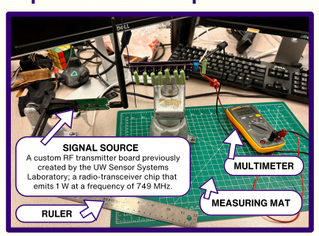
One key component of the rectifier design was the use of the **Schottky diode.** The *Skyworks SMS3924* was found to be the best performing diode for the high-frequency, high-power RF harvesting investigated in this project.

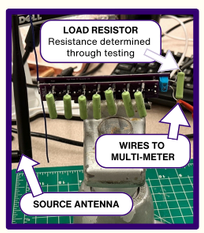
- **LOW FORWARD VOLTAGE DROP**
  - Lose less power, improves efficiency
- **FAST SWITCHING SPEED**
  - Improved efficiency
  - Reduced component size
- **HIGH FREQUENCY CAPABILITIES**
  - Better equipped to receive UHF signals

Building the prototype

- Fabrication and printing of PCBs by **OSH Park**
- Electrical components soldered onto both boards using **solder paste** and a **hot plate reflow technique**
- Connectors and antenna hand-soldered
  - Antenna length: **10 cm** on each side

Experimental set-up





Voltage measured as a function of prototype distance to source

- Measurements began at 1 cm and stopped at 15 cm
- Incremented 0.5 cm after each data point

3 TOTAL TRIALS CONDUCTED

RESULTS

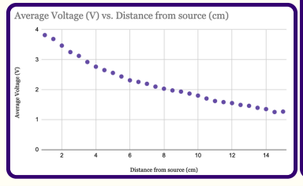
500Ω load resistor was chosen to test with.

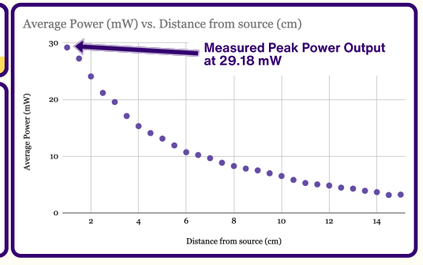
Power calculated from voltage with Ohm's Law

$$P = \frac{V^2}{R}$$

Data was recorded and plotted using Google Sheets.

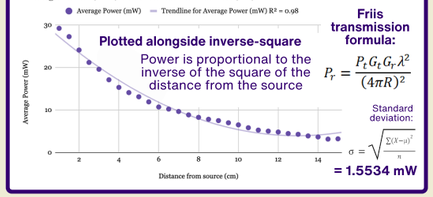
Resistor Value	Voltage Measured	Output Power
1000 Ω	3.9 V	0.01521 W
50 Ω	0.8 V	0.0128 W
500 Ω	***3.4 V***	***0.02312 W***





STATISTICAL ANALYSIS

Peak power output: 29.1848 mW at 1 cm away from source



Friis transmission formula:

$$P_r = \frac{P_t G_t G_r \lambda^2}{(4\pi R)^2}$$

Standard deviation:

$$\sigma = \sqrt{\frac{\sum (X-\mu)^2}{n}}$$

= 1.5534 mW

CONCLUSIONS

In this research, an overview of the design, implementation, and test results of a UHF harvester rectarray prototype and power combiner are presented, and future applications and potential research are introduced. A peak power output of 29.18 mW is achieved from this prototype, demonstrating a **3% efficiency** from its 1W UHF signal source and the ability to harvest high amounts of power. This research shows promising results for the future of energy harvesting, and for sustainable solutions to traditional energy sources.

NEXT STEPS

I am excited about my results, and have the following ideas about future investigations to further explore the prototype harvester board's efficiency in producing DC power output.

Explore Different Harvester Designs	• Utilizing a different number of rectifiers • Different topology/layout
Explore Different Antenna Designs	• Test with an antenna design with a higher directivity, wider bandwidth
Explore a Wider Range of Compatible Frequencies	• Harvesting across entire cellular spectrum (low to high band) broadens potential uses and enhances versatility
Use a High-Power Input Source to Generate More Power Output	• Power input initially limited for safety • Assuming 3% efficiency, could result in >3 W power from high-power input

KEY REFERENCES

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C. R. Valenta and G. D. Durgin, "Harvesting wireless power: Survey of energy-harvester conversion efficiency in far-field, wireless power transfer systems," in IEEE Microwave Magazine, vol. 15 (4), pp. 108–120, June 2014, doi: 10.1109/MMM.2014.2309499.

V. Liu, et al., "Ambient backscatter: wireless communication out of thin air," Proceedings of ACM SIGCOMM, August 2013, pp. 39–50, doi: 10.1145/2486001.2486015.

S. M. Garman, V. Affandy and J. R. Smith, "Harvesting Watts at Ultra-High Frequencies," 2023 IEEE Wireless Power Technology Conference and Expo (WPTCE), San Diego, CA, USA, 2023, pp. 1-6, doi: 10.1109/WPTCE56855.2023.10216166.

Project Bibliography contains full list.

ACKNOWLEDGEMENTS: *I would like to thank the University of Washington Sensor Systems Lab (P.I. Joshua R. Smith) for providing equipment, purchasing the electrical components used in this project, and allowing me to conduct my research in their space.*

Harvesting Power at High Frequencies

This poster won 1st Place, the AFCEA Excellence in Technology award, and the Sigma Xi Best Interdisciplinary Research award at the 2024 Washington State Science and Engineering Fair, and it tests an RF power combiner and rectarray as an alternative to the traditional battery.

Lights, Camera, Action!

Illuminating Los Angeles and Hollywood in the Early 1900s

Maxwell Yu, Saanvi Skanda Subramanian

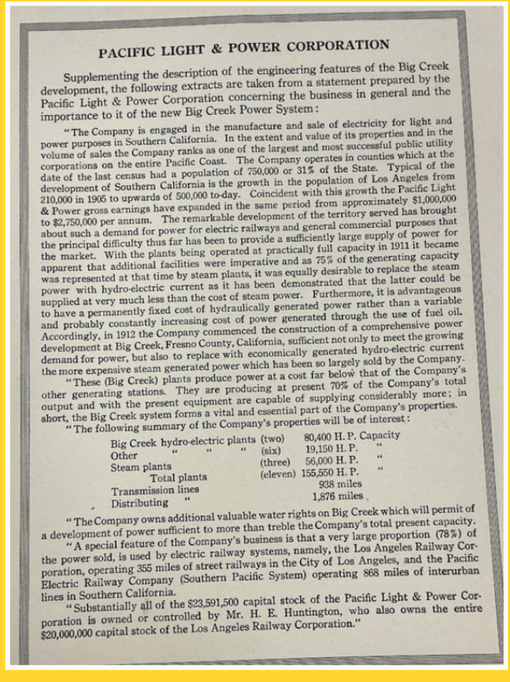
Bishop, G. Haven, Street Lighting Post, 1910

Lindahl, Stanley E., Spherical Street Lighting Wind Tunnel Test Photograph, 1965

Lane, G. S., Heinz 57 Sign, 1916-12-10

Unknown creator, Huxley house with Hollywoodland sign in background, 1940

Parker, Harold A., "Throop students checking the efficiency of the Pasadena Municipal Lighting Plant," around 1913



PACIFIC LIGHT & POWER CORPORATION

Supplementing the description of the engineering features of the Big Creek development, the following extracts are taken from a statement prepared by the Pacific Light & Power Corporation concerning the business in general and the importance to it of the new Big Creek Power System:

"The Company is engaged in the manufacture and sale of electricity for light and power purposes in Southern California. In the extent and value of its properties and in the volume of sales the Company ranks as one of the largest and most successful public utility corporations on the entire Pacific Coast. The Company operates in counties which at the date of the last census had a population of 750,000 or 31% of the State. Typical of the development of Southern California is the growth in the population of Los Angeles from 210,000 in 1905 to upwards of 500,000 to-day. Coincident with this growth the Pacific Light & Power gross earnings have expanded in the same period from approximately $1,000,000 to $2,750,000 per annum. The remarkable development of the territory served has brought about such a demand for power for electric railways and general commercial purposes that the principal difficulty thus far has been to provide a sufficiently large supply of power for the market. With the plants being operated at practically full capacity in 1911 it became apparent that additional facilities were imperative and as 75% of the generating capacity was represented at that time by steam plants, it was equally desirable to replace the steam power with hydro-electric current as it has been demonstrated that the latter could be supplied at very much less than the cost of steam power. Furthermore, it is advantageous to have a permanently fixed cost of hydraulically generated power rather than a variable and probably constantly increasing cost of power generated through the use of fuel oil. Accordingly, in 1912 the Company commenced the construction of a comprehensive power development at Big Creek, Fresno County, California, sufficient not only to meet the growing demand for power, but also to replace with economically generated hydro-electric current the more expensive steam generated power which has been so largely sold by the Company.

"These (Big Creek) plants produce power at a cost far below that of the Company's other generating stations. They are producing at present 70% of the Company's total output and with the present equipment are capable of supplying considerably more; in short, the Big Creek system forms a vital and essential part of the Company's properties.

"The following summary of the Company's properties will be of interest:

Big Creek hydro-electric plants	(two)	80,400 H. P. Capacity
Other " " "	(six)	19,150 H. P. "
Steam plants	(three)	56,000 H. P. "
Total plants	(eleven)	155,550 H. P. "
Transmission lines		938 miles
Distributing "		1,876 miles

"The Company owns additional valuable water rights on Big Creek which will permit of a development of power sufficient to more than treble the Company's total present capacity.

"A special feature of the Company's business is that a very large proportion (78%) of the power sold, is used by electric railway systems, namely, the Los Angeles Railway Corporation, operating 355 miles of street railways in the City of Los Angeles, and the Pacific Electric Railway Company (Southern Pacific System) operating 868 miles of interurban lines in Southern California.

"Substantially all of the $23,591,500 capital stock of the Pacific Light & Power Corporation is owned or controlled by Mr. H. E. Huntington, who also owns the entire $20,000,000 capital stock of the Los Angeles Railway Corporation."

Pacific Light & Power Corporation, 1913

Lighting Los Angeles Exhibit excerpt (Maxwell)

Los Angeles was one of the first cities to adopt electric street lighting (Wuebben 98). Electrical power generation and consumption are closely related, with neither useful without the other. Lighting is one important application that was enabled by the electrification of the LA area, and it complemented other uses of electricity.

This exhibit showcases several moments in the story of lighting up Los Angeles, several of which involve Caltech.

Contextualizing the Heinz 57 Sign with the Los Angeles Railway Corporation: Socioeconomic and Corporate Gain from Technological Advancements (Saanvi)

The years between 1913 to 1945 marked an explosive growth for the greater Los Angeles region, overseeing the birth of modern Hollywood, mass weaponry creation and distribution during the second World War, and the hosting of the city's first major Olympics. With the massive population influx came an immediate need for streamlining electricity throughout the city, spurring the formation of major technological advancements like the Los Angeles Electric Railroad was a game-changer compared to the traditional steam-powered trains of the previous century with increased speed and efficiency. The Huntington Digital Library revealed the enormous and seemingly bizarre Heinz 57 sign originally located in Culver City, created as an advertisement intentionally placed for viewing from the newly constructed Los Angeles Electric Railroad train line. This exhibit aims to contextualize this artifact through exploring the socioeconomic consequences associated with the Los Angeles Electric Railroad.

Lights, Camera, Action!

This history poster with Maxwell Yu uses archival photographs and documents to show how Los Angeles and Hollywood were illuminated in the early 1900s.

Parvaiyattam: Dancing for Sight

ILLUMINATING NAWALAPITIYA'S FUTURE

2023
2024

Prisha Kuberan, Kaustubh Grama, Saanvi Subramanian
Community Giving Project
Bothell High School
9130 NE 180th St, Bothell, WA 98011

April 9th, 2024

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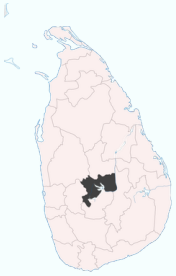
I. EXECUTIVE SUMMARY

DESCRIPTION OF PROBLEM

The Sri Lankan civil war has left smaller communities with heightened poverty rates and socioeconomic struggles, resulting in limited access to healthcare. In the small Tamil town of Nawalapitiya, located in Kandy District, **33%** of the population suffers from cataracts. Cataracts are a condition of cloudiness in the lens of the eye, resulting in a drastic decline in visual function. This condition has been associated with diminished quality of life and even increased physical injury/mortality rates among the elderly. Even though cataract surgery is remarkably safe (**98%** success rate), **67%** of Kandy District Sri Lankan cataract patients reported they were not able to afford the surgery.



OUR SOLUTION

We partnered with the Washington State Sri Lankan Tamil Association (WSSTA) to put together 2 performances of Bharatanatyam, a Tamil cultural dance, in an event called **Parvaiyattam** - Tamil for *Dancing for Sight*. All proceeds went to the Nawalapitiya General Hospital to purchase cataract surgical sets.

TARGET AUDIENCE

DEMOGRAPHICS:

- Tamils - South Asian ethnolinguistic group
  - Family audiences
- Age: 20-60 y.o.

PSYCHOGRAPHICS:

- Enjoy Tamil culture
- Interested in Bharatanatyam

GEOGRAPHICS:

- Greater Seattle Area, suburban locations

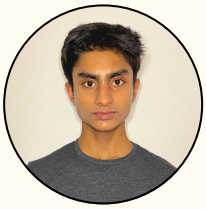
PROJECT GOALS

PURPOSE: To raise funds to aid the people of Nawalapitiya and restore their vision

1. **Raise $10,000 to fund cataract surgical sets in Nawalapitiya**
2. **Raise awareness about healthcare inequalities in Sri Lanka**

OUR TEAM

Prisha Kuberan
PROJECT LEAD
DANCE COORDINATOR



Kaustubh Grama
PROJECT LEAD
LOGISTICS COORDINATOR

Saanvi Subramanian
PROJECT LEAD
MARKETING COORDINATOR

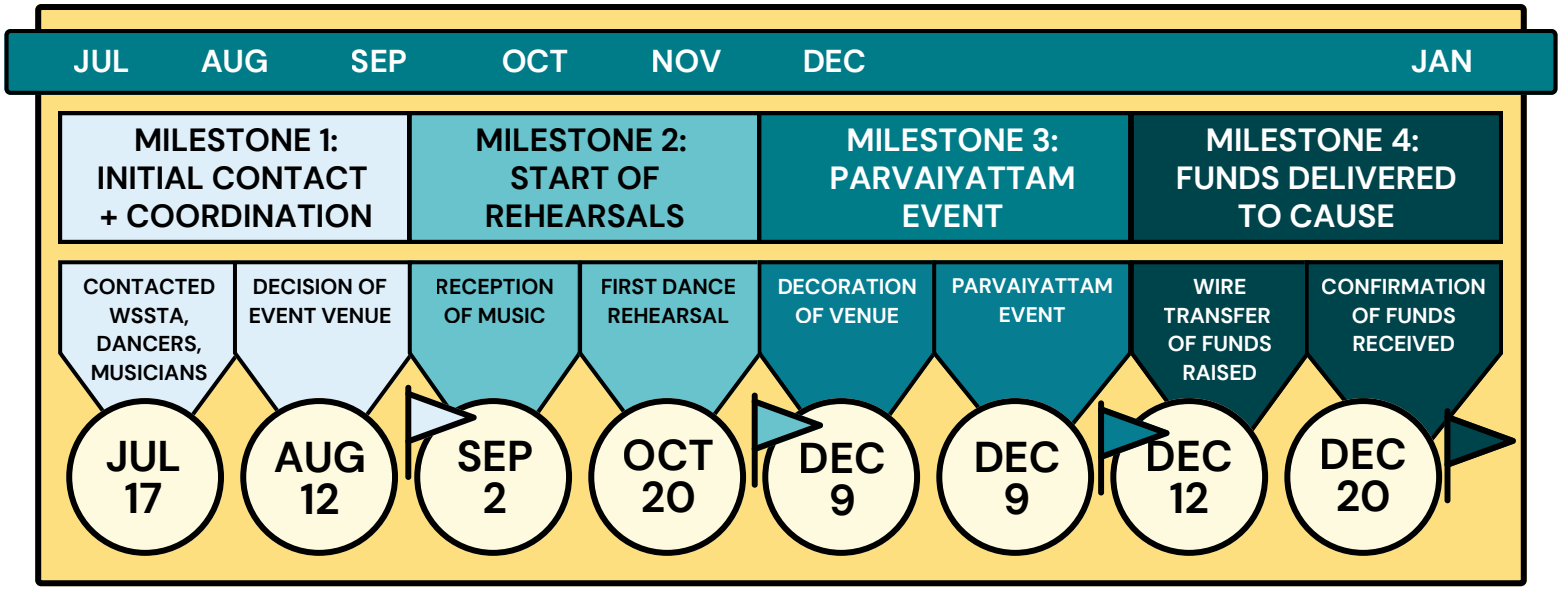
Pratheeba Satkunanathan
WSSTA PRESIDENT

Nrityalaya Dancers
BHARATANATYAM PERFORMERS

Carnatic Musicians
MUSICIAN GROUP

I. EXECUTIVE SUMMARY

SCHEDULE AND MILESTONES



QUALITY MANAGEMENT

Schedule

- ★ Created schedule checkpoints
- ★ ***KEY METRIC: Adherence to milestones (OTD)***

Financials

- ★ Monthly budget reviews with team
- ★ ***KEY METRIC: Budget variance***

Logistics

- ★ Conducted checks on venue cleanliness, seating availability, preparation of back-up equipment
- ★ Sent out a CSAT survey to attendees
- ★ ***KEY METRIC: CSAT score***

Mission Objectives

- ★ Reach out to understand our impact
- ★ ***KEY METRIC: Total funds raised, number of eye surgeries funded***

To ensure our project operated smoothly, we also tracked the **number of attendees** and **number of ticket sales**.

RISK MANAGEMENT

RISK	RESPONSE STRATEGY
Low Number of Attendees *Likelihood: LOW*	• Market with different channels • Advertise to local community • Offer free tickets for children • Offer tickets at the door
Injury to Performers *Likelihood: MODERATE*	• Implement safety protocols during rehearsals • Secure fill-in dancers from the Nrityalaya School of Dance
Falling Short of Donation Goals *Likelihood: HIGH*	• Monitor donation progress • Encourage additional donations from event attendees • Promote cause through posters

BUDGET

Item	Cost per Item	#	TOTAL COST
Lamp Decor	Donated	1	**$0.00**
Flower Decor	Donated	1	**$0.00**
Outfits	$20.50	10	**$205.00**
Venue	$100.00/hr	11 hrs	**$1100.00**
Musicians	$500.00	1	**$500.00**
Total Budget			**$1,805.00**

I. EXECUTIVE SUMMARY

EXECUTION AND OUTCOMES

MILESTONE 1: INITIAL CONTACT AND COORDINATION

Kaustubh contacted WSSTA, the musician ensemble, and scouted dancers with Prisha to form the team.

MILESTONE 2: START OF REHEARSALS

After reserving the venue, Prisha choreographed the dance. She created a rehearsal schedule with the dancers, and held the first Parvaiyattam rehearsal.

ADVERTISING AND TICKET SELLING

Saanvi designed flyers and distributed them to South Asian centers. She sold tickets online using *Sulekha*.

MILESTONE 3: PARVAIYATTAM EVENT

The two Bharatanatyam dance productions of Parvaiyattam took place on **December 9, 2023**. The performances began with a talk from the WSSTA president about the injustices in Sri Lanka, and encouraged donations from attendees.

Renton IKEA Performing Arts Center

3:00 PM - 6:30 PM

7:00 PM - 10:30 PM

MILESTONE 4: DONATION OF FUNDS RAISED

We received confirmation from the Nawalapitiya General Hospital that our donation wire transfer was successful.

EVALUATION OF KEY METRICS

KEY METRIC	*STATUS*	*KEY METRIC*	*STATUS*
Adherence to milestones	⚠	CSAT score	✓
Budget variance	✓	Total funds raised	✓
# of ticket sales	✓	# of eye surgeries funded	✓
# of attendees	✓		

812 TOTAL NUMBER OF ATTENDEES

$22,568 TOTAL AMOUNT RAISED

423 CATARACT SURGERY SETS FUNDED

MONITORING

Created schedule checkpoints for each milestone.

Held budget reviews, tracked expenses with Excel.

Implemented a quality management plan to ensure project operated smoothly, and audience satisfaction of event.

CONTROLLING

- One dancer was injured
- Low amount of ticket sales
- Poor dance synchronization
- Secured alternate dancer to fill in
- Advertised to local community centers
- Held team-bonding activities for dancers

LESSONS LEARNED

- Plan for delays in international communication
- Should become aware of opportunities for nonprofits to increase impact
- Target a broader audience

RECOMMENDATIONS FOR FUTURE PROJECTS

- Research community programs
- Expand target demographics
- Utilize more social media channels

II. INITIATING

A. Statement of the problem

The lasting effects of the Sri Lankan civil conflict have left **Nawalapitiya**, a small suburban town in **Kandy District, Sri Lanka**, with challenges of limited healthcare access and heightened poverty rates. The community struggles with eye health, particularly with regard to cataracts.

Cataract remains the leading cause of blindness. The *National Eye Institute* describes cataracts as cloudy areas in the lens of the eye that significantly affect day-to-day life. Examples include limitations in mobility, activities of daily living, and physical performance. Blindness and visual impairment are not only health issues for the elderly but also significantly affect the quality of their lives.

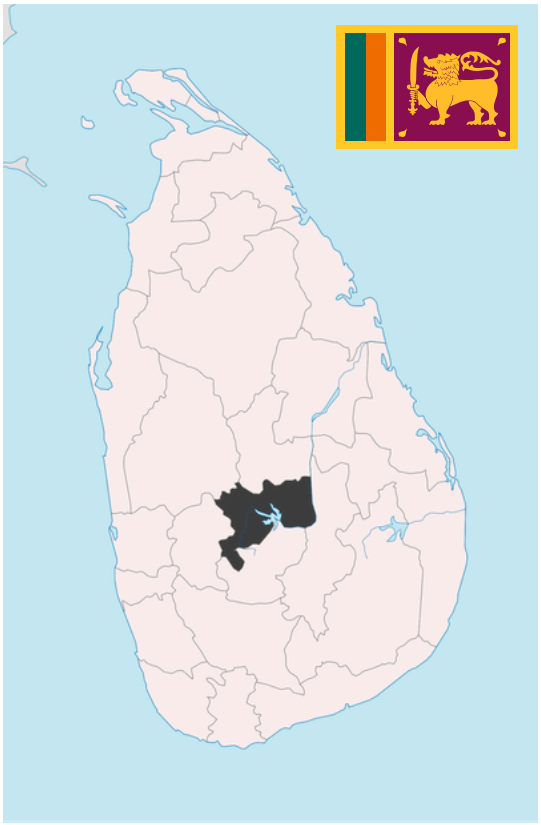
In Kandy District, an estimated **33%** of people experience impaired vision from complications of cataracts. In a *National Library of Medicine* (NLM) paper, researchers found that out of 470 elders (people over the age of 60), 379 (or 80.6%) were found to have some kind of cataract. This study demonstrated lack of awareness and knowledge, socioeconomic factors, and misconceptions as the main barriers to cataract treatment, leading to a lower cataract surgery rate irrespective of the high cataract prevalence reported. Findings of this study highlight the importance of cataracts as a common health problem in the elderly and recommend eliminating barriers to its treatment. However, it is reported that **67%** of Kandy district cataract patients cannot afford treatment.

33%

KANDY DISTRICT POPULATION AFFECTED BY CATARACT

67%

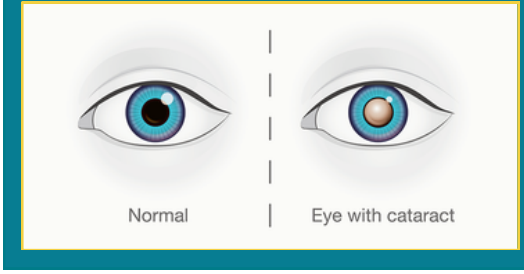
CATARACT PATIENTS CANNOT AFFORD SURGERY



Map showing Kandy District, Sri Lanka

CATARACTS

- Cloudy area in the lens of eye
- Make vision blurry, hazy, or less colorful
- Decrease quality of life
- Can only be cured through surgery



Cataracts are curable, but only with cataract eye surgery. This surgery is safe and effective, helping 98% of people with no complications. Unfortunately, **the price of cataract eye surgery is simply unaffordable for Nawalapitiya residents.** With Parvaiyattam, we aim to make a difference in the community by making better cataract surgery treatments available and helping those affected regain their vision to lead happier, healthier lives.

98%

CATARACT SURGERY SUCCESS RATE

B. Project scope

Description of project

This project aimed to deliver funds to the Nawalapitiya community to provide cataract surgical sets for those affected by the condition. Towards this, we created a fundraising project by partnering with WSSTA - the Washington State Sri Lankan Tamil Association - to put together two performances of Bharatanatyam, a Tamil cultural dance, in an event called **Parvaiyattam** (*Dancing for Sight)*. Our goals were 1) to raise funds by selling tickets for the event and encouraging donations from attendees for the Nawalapitiya General Hospital and 2) to create awareness about the healthcare inequalities in Sri Lanka. The project comprised the planning and logistics for the event, campaigns through organic and social media reach, performance, and delivery of funds to provide access to eye care for the community.

Bharatanatyam - a Tamil cultural dance performed with a group of trained dancers in cultural attire

Nawalapitiya General Hospital - a district hospital in Sri Lanka providing healthcare services to 3,000+ people

TARGET AUDIENCE

DEMOGRAPHICS:
- Tamil families
- South Asians
- Age: 20-60 y.o.

GEOGRAPHICS:
- Greater Seattle Area
- Suburban locations

PSYCHOGRAPHICS:
- Enjoy Tamil culture
- Interested in Bharatanatyam

PROBLEM IDENTIFIED

Civil conflict in Sri Lanka created socioeconomic difficulties and healthcare challenges for small, Tamil towns, specifically with cataract

OUR CAUSE

Help fund cataract surgical equipment in Nawalapitiya

OUR PROJECT

Parvaiyattam event:
2 Tamil classical dance performances to raise money and awareness to aid Nawalapitiya

PARTNERING NONPROFIT

WSSTA - Washington State Sri Lankan Tamil Association

வாஷிங்டன் மாநில இலங்கை தமிழ்ச் சங்கம்

WASHINGTON STATE SRI LANKAN TAMIL ASSOCIATION

Purpose

Parvaiyattam focused on raising funds (goal: $10,000) for the Nawalapitiya General Hospital in collaboration with WSSTA. With these funds, 187 cataract surgery equipment sets could be distributed. Cataract surgery can last as little as 10 minutes, but the benefits are worthwhile for a lifetime of clear vision. It allows people to reconnect visually, socially, and emotionally with their loved ones, improving their quality of life. It also significantly reduces the chance of severe physical injury in elders, reducing future treatment costs. By making cataract surgery a possibility for the people of Nawalapitiya, we can lead them to a brighter and more sustainable future.

Rationale

Tamils are an ethno-linguistic group located in South Asia. Hearing about the devastating plight of Tamils in Sri Lanka following their Civil War, we reached out to the Washington State Sri Lankan Tamil Association (WSSTA) to understand more about the issue and how we could help. We learned about the limited access to healthcare and how it disproportionately affects small communities. Higher costs are associated with acquiring healthcare necessities and procedures for those in rural areas, as there is high demand in contrast to limited access. Unfortunately, rural/small-town populations cannot afford treatment, even for relatively simple concerns, such as cataracts.

We identified the small town of Nawalapitiya, located in Kandy District, which currently struggles with limited healthcare access, most prevalently regarding cataracts. Cataracts are the clouding in the clear eye lens, often drastically inhibiting the patient's field of vision and affecting 33% of Kandy District. Cataracts are associated with subsequent physical disability and have even been correlated to increased mortality. Regardless of the cause, decreased visual function is associated with diminished quality of life and general function.

Fortunately, cataract surgery is a remarkably safe outpatient procedure that rarely produces complications, boasting a 98% success rate. Cataract surgery causes significant improvement in visual function and cognitive, emotional, and general well-being. Unfortunately, a survey published by the National Institute of Health reveals that a staggering 67% of those affected by cataracts in Kandy District responded "no" when asked if they were able to pay for surgery.

Upon learning about the debilitating effects of cataracts, we wanted to create a project to assist Nawalapitiya's journey to clear vision and a brighter future. **We wanted our project to involve Tamil culture to appeal to the Tamil and South Asian communities of the greater Seattle area in order to attract people who would be most inclined to support the cause and enjoy Bharatanatyam.**

Expected Benefits

From the Parvaiyattam project, we can expect the following benefits:

- **Raise funds for cataract surgical equipment in Nawalapitiya**
- **Promote Tamil culture through cultural dance and connect with community**
- **Raise awareness about socioeconomic and healthcare inequalities in Sri Lanka**

III. PLANNING AND ORGANIZING

A. Project goals

Primary - Finance: Raise $10,000 for cataract surgical equipment

The primary goal of the Parvaiyattam project was to raise enough money to help alleviate the costs associated with **187** cataract surgeries in Nawalapitiya, given the calculated cost of **a set of one-time use surgical equipment** is $53.31 (USD). The breakdown of this cost is as follows: $2.17 for a phaco knife (used for clear corneal incision (CCI), near corneal incisions and scleral tunnel incisions in cataract surgery), $4.65 for vision blue dye (a sterile solution used for medical aid), $9.30 for methyl cellulose (used to relieve dry, irritated eyes), and $37.19 for a pair of lenses to finish the cataract surgery.

Item	Price in rupees	Price in dollars
Phaco knife	Rs. 700	$2.17
Vision blue dye	Rs. 1,500	$4.65
Methyl cellulose	Rs. 3,000	$9.30
Lenses	Rs. 12,000	$37.19
Total	Rs. 17,200	$53.31

A cataract surgical set

These funds were raised by selling tickets for the event - $20 tickets, with free entry for children under the age of 10. To meet this goal, we planned to sell 450 tickets to the event and raise $1,000 by encouraging additional donations from event attendees.

***TARGET**: $9,000 from ticket sales, $1,000 from donations*

Secondary - Community Awareness: Raise awareness about the significant impacts of healthcare inequalities in Sri Lanka

The secondary objective was to raise awareness about the limited healthcare access in Nawalapitiya and their struggles with cataract-related vision impairment. We did this by opening the Parvaiyattam event with a speech from the WSSTA president about the issues that Sri Lanka is facing due to civil conflict, specifically regarding to healthcare inequalities in smaller cities. This topic was the main focus of the event, with detailed posters put in place to explain the problem and demonstrate the impact of any donations on the lives of those in Nawalapitiya. By raising community awareness, especially within the South Asian community, we can impart positive change to Nawalapitiya and help countless people see the world better.

B. Human resource management plan

Our team consisted of three DECA members who organized and ran the Parvaiyattam event, the WSSTA President, and the dancers and musicians who partook in the productions.

Our Team:

PRISHA KUBERAN *– Project Lead and Dance Coordinator*

Responsibilities: Leading the project, choreographing dance routines, coordinating rehearsals, initiating contact with WSSTA, and ensuring the execution of the entire initiative.

Strengths and skills: Strong leadership, project management, communication.

Prisha organized the dance productions, serving as the choreographer and coordinator for the group of dancers. Additionally, she initiated contact with WSSTA.

KAUSTUBH GRAMA *– Project Lead and Logistics Coordinator*

Responsibilities: Managing finances, contacting external contributors, finding a suitable venue, and overseeing strategic planning to ensure the success of the initiative.

Strengths and skills: Financial acumen, strategic planning, communication.

Kaustubh led the logistics, strategic planning, and financial aspects of the project. He managed finances and played a crucial role in planning, including contacting singers in India and securing an appropriate venue.

SAANVI SUBRAMANIAN *– Project Lead and Marketing Coordinator*

Responsibilities: Designing posters, finding a ticket selling platform and enabling its use, connecting with local temples and community organizations, sharing event through online platforms.

Strengths and skills: Social media, graphic design, event marketing, and advertisement.

Saanvi advertised the Parvaiyattam event and sold tickets through *Sulekha*, an online platform. She designed posters to spread the word and shared them with local organizations and online social media platforms including Whatsapp and Facebook.

PRATHEEBA SATKUNANATHAN *– WSSTA President and Sponsor of Event*

Responsibilities: Connecting with the hospital, securing funding, fostering collaboration, and advocating for the project within the wider community of Sri Lankan Tamils.

Strengths and skills: Networking, community engagement.

Pratheeba Satkunanathan provided financial support for the initiative and facilitated collaboration between the project and the hospital.

BHARATANATYAM DANCERS *Varshni Iyer, Joshitha Velmurugan, Adthvika Charu, Akchara Arun, Meghna Sekaran, Kanika Saravanan, Thanushri & Krithi Narandren, Bavishya Rajasekar, Prisha Kuberan*

Responsibilities: Learning and performing dance routines, actively participating in rehearsals, and contributing to the artistic success of the Bharatanatyam productions.

Strengths and skills: Bharatanatyam dancing, teamwork, commitment.

These 10 dancers contributed to the artistic success of the project through commitment, teamwork, and active involvement in rehearsals and performances.

CARNATIC MUSICIANS *N. K. Srividya – Vocal, R. L. Srivatsa – Flute, B. T. Mayan – Percussion, T. N. Rajesh – Rhythm Pad*

Responsibilities: Rehearse together and audio recordings of requested songs.

Strengths and skills: South Asian classical music, experience with Bharatanatyam music.

Located in Chennai, India, these musicians contributed to the musical success of the production through their expertise in Carnatic classical music and accompanying the Bharatanatyam dancers.

C. Schedule

i. Milestone

MILESTONE 1

Initial Contact and Coordination

MILESTONE 2

Start of Rehearsals

MILESTONE 3

Parvaiyattam Event

MILESTONE 4

Funds Delivered to Cause

MILESTONE 1: INITIAL CONTACT AND COORDINATION

Description: The first milestone of the project comprises the following activities: contacting WSSTA to initiate collaboration and confirm project feasibility, coordinating with the Nrityalaya School of Dance dancers, reaching out to the musicians, and assessing the venue - leading to identifying and reserving the Renton IKEA Performing Arts Center. During this phase, we also defined our goals for the project, identified resources, and prepared a high-level schedule. This milestone marks the completion of the initial stages of the project. Once we received confirmation from the involved organizations, we were able to begin preparations for the Parvaiyattam event.

Activities: Initial discussions, project proposal, communication with WSSTA and hospital representatives, venue reservation	**Outcome:** Confirmation of collaboration, project planning and viability, venue reservation

MILESTONE 2: START OF REHEARSALS

Description: With the musical arrangements in place, rehearsals for the Bharatanatyam production commenced. We formed a team of dancers and recruited musicians to help prepare for the performance. This included separate dance and music rehearsals, with dress rehearsals leading to the event date. Dance rehearsals were at Nrityalaya School of Dance in Redmond, WA, and musician rehearsals were at their residences in Chennai, India. Dance practices were planned to be intense for at least 15 hours a week in the 2 months leading up to the performance.

Activities: Receiving music recordings, choreography sessions, practice schedules, coordination with dancers	**Outcome:** Music recordings finalized, Initiation of dance rehearsals, ensuring preparation for the upcoming production

MILESTONE 3: PARVAIYATTAM EVENT

Description: The Parvaiyattam event was planned for Saturday, December 9th at the Renton IKEA Performing Arts Center, and consisted of two complete productions: the first from 3:00 pm to 6:30 pm, and the second from 7:00 pm to 10:30 pm. This was done to maximize contributors' attendance and optimize our total funds raised. Both productions were held on the same day to minimize venue rental costs. Each paying attendee contributed $20 in ticket sales, while children under 10 were admitted free of cost. Throughout the event, we invited speakers from WSSTA to speak about Tamils in Sri Lanka and the limited access to healthcare in rural areas, helping us achieve our secondary goal of raising awareness for our cause.

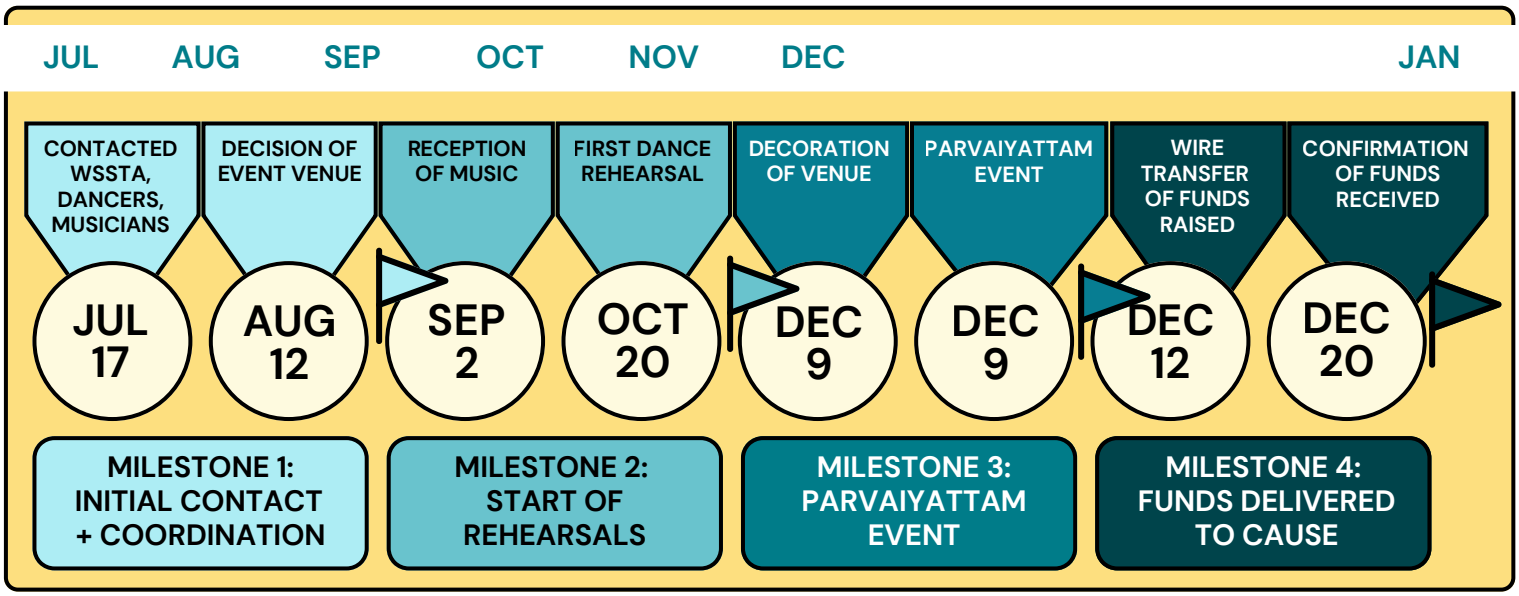
Activities: Decorating the venue, two showings of Parvaiyattam dance performances, posters and talks on cause	**Outcome:** Funds raised in ticket sales, increased awareness about healthcare inequities in Sri Lanka

MILESTONE 4: FUNDS DELIVERED TO CAUSE

Description: Following the completion of the Parvaiyattam event, we formally presented the Washington State Sri Lankan Tamil Association with the money raised from the project. The money was then sent to the director of the Nawalapitiya General Hospital to purchase cataract surgery equipment. We confirmed the funds were successfully received by the Nawalapitiya General Hospital through contact via email.

Activities: Wire transfer of collected funds from WSSTA to Nawalapitiya General Hospital, confirm money transfer	**Outcome:** Provide access to cataract surgery for residents of Nawalapitiya, donate funds raised, ensure quality

ii. Timeline to reach each milestone



D. Quality management plan

We created a holistic quality plan to ensure that our project operated smoothly. This comprised addressing the desired outcomes on our **schedule, financials, logistics**, and **mission objectives**. Quality management includes defining the quality standards, managing the control parameters, and a plan validating the event's quality objectives. Communication channels were established, ensuring quality issues were addressed in the earliest time. We identified the following quality metrics, along with general metrics of **number of attendees** and **number of ticket sales.**

SCHEDULE

Ensuring the schedule involves having mechanisms that ensure the timeliness of event parameters. We will schedule regular checkpoints to validate the venue's availability and preparations leading to the event. Validations on preparations include signage at the event, setting up the event hall, marking the parking areas, and ensuring the availability of parking access. We set up additional checkpoints on weather conditions and potential events in the neighborhood to prepare for contingencies.

KEY METRICS: Adherence to milestones (On-time delivery)

FINANCIALS

We worked within strict constraints on the allocated budget. We set up monthly reviews on cost incurrence and deviations from the budget. We also recognized the need to ensure a safe, trusted way of receiving donations from our patrons. We diligently reviewed vendor products to identify the most reliable means for donations that comprehensively address privacy and identity risks. Our chosen vendor met industry and regulatory standards for financial certifications.

KEY METRICS: Budget variance

LOGISTICS

Our team conducted rigorous checks to ensure the program ran smoothly and that the performers and attendees had a great experience. To this end, we set up an extensive list of event logistics—availability of seating, cleanliness of event space, and setting up backup microphones—to name a few. We will also send out a post-event survey to attendees to measure their satisfaction with the event.

KEY METRICS: CSAT score

MISSION OBJECTIVES

Finally, we defined quality objectives for the impact of the entire project, which ultimately was the number of people treated from the funds raised. We ensured successful transfer of the funds, and measured impact from the donations to confirm that the funds delivered the intended objective.

KEY METRICS: Total funds raised, number of eye surgeries funded

E. Risk management plan

RISK	IMPACT	RESPONSE STRATEGY
Low Number of Attendees *Likelihood: LOW*	**Detrimental** • Reduced fundraising success • Diminished community involvement	• Enhanced marketing and promotion through different channels • Advertise to local community • Offer free tickets for children • Offer tickets at the door
Injury to Performers *Likelihood: MODERATE*	**Detrimental** • Production severely impacted through the absence of a part of the performance	• Implement strict safety protocols during rehearsals • Maintain a pool of back-up dancers from the Nrityalaya School of Dance to fill in
Falling Short of Donation Objectives *Likelihood: HIGH*	**Detrimental** • Unable to meet project goals ◦ Reduced number of surgical sets funded for	• Continuously monitor donation progress • Encourage additional donations from event attendees • Further promote cause through posters

F. Proposed project budget

Item	Cost per Item	#	TOTAL COST
Lamp Decor	In-Kind from Nrityalaya	1	**$0.00**
Flower Decor	In-Kind from Nrityalaya	1	**$0.00**
Bharatanatyam Outfit	$20.50	10	**$205.00**
Renton IKEA Rental	$100.00/hour	11 hrs	**$1100.00**
Accompaniment Group	$500.00	1	**$500.00**
GRAND TOTAL			**$1,805.00**

WSSTA agreed to cover a maximum of $4,000.00 for the costs associated with the program. Being mindful of their contribution, we wanted to minimize our costs and take advantage of donations. Nrityalaya School of Dance provided production decorations, most of which were recycled from prior events. Additionally, we were able to secure microphones and audio equipment free of cost from the Renton IKEA Performing Arts Center. Altogether, we projected a budget of **$1,805.00** for the Parvaiyattam event.

IV. EXECUTION

MILESTONE 1: INITIAL CONTACT AND COORDINATION

We initially coordinated with the Washington State Sri Lankan Tamil Association (WSSTA), Nrityalaya School of Dance, accompanying musicians, and the Renton IKEA Performing Arts Center for the project. We contacted the president of WSSTA to finalize our idea of holding Bharatanatyam performances in an event, and she connected us with representatives from Nawalapitiya General Hospital. We worked with Nrityalaya to scout our 10 leading dancers, including Prisha, as well as six emergency replacement dancers. We reached out to 4 Carnatic musicians in Chennai, India, who had connections with Nrityalaya: Srividya for singing, Srivatsa on percussion, Mayan on flute, and Rajesh on rhythm pad. After identifying our team, we confirmed our date with the Renton IKEA Performing Arts Center and proceeded with operations accordingly.

Renton IKEA Performing Arts Center

வாஷிங்டன் மாநில
இலங்கை தமிழ்ச் சங்கம்

WASHINGTON STATE
SRI LANKAN TAMIL ASSOCIATION

MILESTONE 2: START OF REHEARSALS

Kaustubh coordinated with the group of artists and sent them a list of the 9 songs needed for 2 productions. Kaustubh asked that the recordings be sent on September 2nd so that the dancers and Nrityalaya teachers would have adequate time to rehearse with the music.

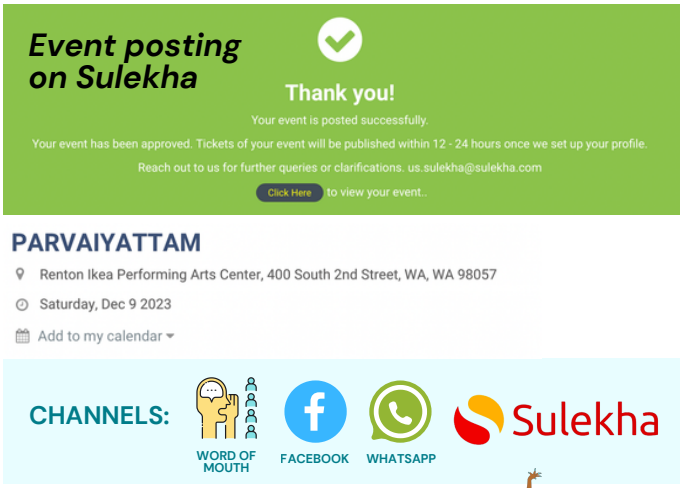
Prisha was in charge of managing the dance rehearsals and choreography of the 9 pieces. Before rehearsals began, Prisha set a rigorous practice regimen for each week leading up to the production, lasting 2 months. Prisha coordinated with the dancers to set practice times leading up to the event. She requested that everyone enter their availabilities into a shared Excel spreadsheet so that she could get a sense of what times work for the dancers. Once everyone populated the spreadsheet, she chose the least occupied times, and the dancers rearranged their commitments accordingly.

Name	Day	Please note that the finalized schedule will be for the next three months!					
	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday	Sunday
Prisha	5:00 pm - 9:00pm	NONE	5:00-9:30	NONE	5:00-9:00	7:00-12:00	7:00-12:00
Varshini	NONE	5:00- 7:00	5:00-9:30	NONE	NONE	ALL DAY	ALL DAY
Karishma	1:00 pm - 7 :00pm	NONE	5:00-9:30	NONE	1:00-7:00	9:00-10:00	9:00-10:00
Akchara	7:00 pm- 9:00pm	6:00-7:00	5:00-9:30	7:00-9:00	7:00-9:00	8:00-11:00	8:00-11:00
Tanu	5:00 pm - 9:00 pm	7:00-9:00	5:00-9:30	NONE	5:00-9:00	9:00-12:00	9:00-12:00
Krithi	5:00pm- 9 pm	NONE	5:00- 7:00	NONE	5:00-9:00	ALL DAY	ALL DAY
Advika	NONE	6:00-9:00	5:00-9:30	5:00-9:00	NONE	9:00-12:00	9:00-12:00
Joshitha	5:00pm - 7:00pm	NONE	5:00-8:45	NONE	5:00-7:00	7:00-9:00	7:00-9:00
Kanika	7:00 pm- 9:00pm	6:00-9:00	5:00-9:30	5:00-8:00	7:00-9:00	8:00-12:00	8:00-12:00
Bavishya	6:00 pm- 9:00pm	5:00-9:00	5:00-9:30	NONE	6:00-9:00	7:00-9:00	7:00-9:00
Finalized Schedule:		**Please confirm this by marking the boxes with your color!**					
Monday - 5:00pm - 8:30 pm							
Wednesday- 5:00pm - 8:30pm							
Friday - 5:00pm - 8:30pm							
Saturday -9:00am - 12:00pm							
Sunday - 9:00 am - 12:00 pm							

ADVERTISING AND TICKET SELLING

Saanvi was in charge of advertising the Parvaiyattam event and finding and setting up the ticket-selling platform. She designed the flyers shown to the right. She distributed them to Hindu temples and Tamil schools around the greater Seattle area to attract the South Asian community and local businesses willing to promote the event. She also realized that digital advertising would be effective, as it could better target South Asian people who may be interested in this cultural event and allow for easier spreading of the event information. She sent out digital copies of the flyer on platforms such as Facebook and WhatsApp to attract family audiences of South Asians and others who may be interested in learning more about its culture.

Saanvi chose to sell tickets for the event through the platform *Sulekha*, an online platform dedicated to South Asian community events. She thought it would be a perfect platform for Parvaiyattam and another channel by which we could advertise the event. Sulekha also does not charge a commission fee, so no operating costs were associated with the platform, another driving factor contributing to our choice. We also offered the opportunity to purchase tickets at the door.



MILESTONE 3: PARVAIYATTAM EVENT

Parvaiyattam was held on Saturday, December 9th, 2023, at Renton IKEA Performing Arts Center. The event consisted of two complete productions: the first from 3:00 pm to 6:30 pm and the second from 7:00 pm to 10:30 pm. This was done to maximize contributors' attendance and optimize our total funds raised. Both productions were held on the same day to minimize venue rental costs.

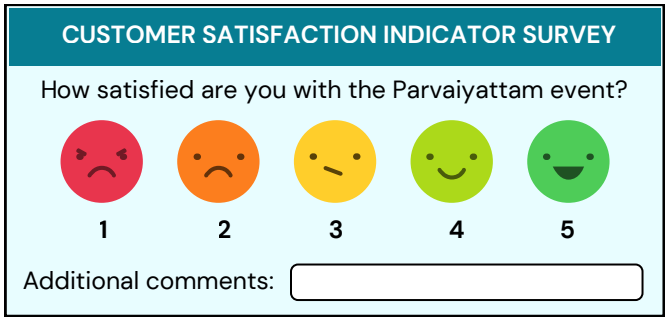
Before attendees arrived, the dancers completed rehearsal from 12:00 pm to 2:30 pm. During this time, the parents of the dancers volunteered to decorate and prepare the venue for guests. They also set up a ticket-selling booth to allow for on-the-spot sales, set up next to our posters that advertised our cause.

Before the two performances began, we invited Pratheeba Satkunanathan and a couple of other WSSTA officials to talk about the socioeconomic injustices of Tamils in Sri Lanka, and the limited access to healthcare in rural areas. This helped us achieve our secondary goal of raising awareness for our cause.

10 Bharatanatyam dancers performed in the event, and we had **812** total guests at the venue (**337** and **475** in the first and second showings, respectively), out of which were **501** paying attendees - **299** in the first performance and **202** in the second performance. We raised **$10,020** through organic ticket sales and **$1,264** from additional donations, for a total of **$11,284** raised.

The dancers performed 9 pieces showcasing Tamil culture, depicting episodes from the Sri Lankan civil war, and how it affected Sri Lankan Tamils. The pieces were set to music recordings provided by our accompaniment team in Chennai.

We printed out QR codes that when scanned, directed guests to a short, online survey assessing attendee satisfaction. At the end of each program, we thanked the attendees for coming and encouraged them to scan the QR codes to provide feedback from the event.



MILESTONE 4: FUNDS DELIVERED TO CAUSE

Our team met with Pratheeba Satkunanathan (WSSTA President) on December 11, 2023, to discuss the next steps. The money we collected in ticket sales and donations was held in Saanvi's personal bank account linked with Sulekha. When we met with Pratheeba, we found out about the Microsoft Matching Gifts Program, and she provided us with the contact information of a WSSTA member whom we could contact to help us with the process. Once we got Microsoft to match our funds, we transferred the money to the WSSTA bank account. On December 23, we got confirmation from the Nawalapitiya General Hospital's Director, Maduwantha Dissanayake, that our wire transfer was successful. The order of 423 sets of cataract equipment was placed on December 28, 2023; as of April 5th, they have been used to aid in 102 surgeries.

MICROSOFT MATCHING GIFTS PROGRAM

Doubling the Donation

In our efforts to maximize our impact, we continued to look for more ways to increase our funds after the event. After discussing various options with WSSTA and the Parvaiyattam team, we stumbled across the Microsoft Matching Gifts Program. The Microsoft Matching Gifts Program is a corporate initiative encouraging employees to donate to eligible non-profits. Microsoft matches donations, effectively doubling the impact of their contributions, showcasing the company's commitment to corporate social responsibility. Leveraging this unique opportunity, we sought out Microsoft employees who were willing to aid in our impact. We met with a WSSTA member who was a Microsoft employee and had participated in the company's matching program before. We then transferred the funds to this trusted personnel, and applied to the program, designating the donation for WSSTA. Our application was successful, and by presenting the contribution as a charitable donation to a non-profit, **Microsoft matched our amount, effectively doubling the impact of our fundraising efforts.**

EVALUATING OUR IMPACT

The **$22,568.00** raised through our efforts will fund cataract surgery equipment sets for **423** people to help cure their cataracts, greatly exceeding our original goals of raising ten thousand dollars to help 187 patients! From the time of donation to April 5th, the Nawalapitiya General Hospital was able to complete 102 successful cataract surgeries, as depicted in the photos below. These people are now able to connect with their loved ones, contribute to their community, and live safer, more fulfilling lives.

501
PAYING ATTENDEES

$22.5K
TOTAL MONEY RAISED

423
SURGERY SETS PURCHASED

102
PATIENTS HELPED SO FAR

V. MONITORING AND CONTROLLING

A. Monitoring

Schedule

To monitor the schedule of the project, we established checkpoints to ensure we were on track to achieve our milestones. By utilizing the checkpoints below, we were able to organize dates for what needs to be completed and the process by which to address the task best. We tracked our adherence to milestones to assess our on time delivery. This helped us hold ourselves accountable for the completion of tasks needed to ensure the quality of the event.

SCHEDULE CHECKPOINT	DATE STARTED	PROCESS COMPLETED	DATE COMPLETED
Establish Collaboration with WSSTA ☑	July 10, 2023	Contacted WSSTA President about cause and finalized idea for event	July 13, 2023
Artist Coordination ☑	July 14, 2023	Scouted dancers and confirmed collaboration with accompaniment team	July 17, 2023
Event Planning (Time, Location, Date, Concept) ☑	July 17, 2023	Connected with venue and booked Renton IKEA Performing Arts Center	August 12, 2023
Music Recordings ☑	July 17, 2023	Requested and received Accompaniment team's music recordings for event	September 2nd, 2023
Advertising Parvaiyattam ☑	September 2nd, 2023	Distributed flyers to community organizations, through digital channels and a ticketing platform	December 8th, 2023
Day of Event ☑	December 9th, 2023	Parvaiyattam production set up and execution	December 9th, 2023
Funds Delivered ☑	December 11th, 2023	Worked with WSSTA to send funds to Nawalapitiya General Hospital	December 23rd, 2023 ***Checkpoint delayed**

Budget

WSSTA agreed to cover the costs of the Parvaiyattam event, as long as costs did not exceed $4,000. We monitored the event's costs through maintaining an Excel spreadsheet and holding monthly budget check-ins with the Parvaiyattam team during the planning process to go over the costs for the event. After the event, we evaluated the project's financials, as shown below.

Initial investment: $1,805.00
Total revenue: $22,568.00

Return on Investment: $20,763 ⇨ ***1150% ROI***

Quality

To ensure our project quality was held to the highest standards, we created a holistic quality management plan. This plan was in effect for the entirety of project timeline and closely tracked through the execution until the funds were delivered and impact was validated. Throughout the project, we informed the team of issues and responses. Checkpoints were proactively tracked, and identified issues were mitigated immediately. Our rigor on quality management was critical to the success of the program.

B. Controlling

As we prepared for the Parvaiyattam event, we encountered a few issues. While these issues initially posed a risk, by planning for them in our risk management phase, we were able to address them in a timely and orderly manner. These issues did not result in any significant impact to Parvaiyattam, and we were able to execute the project despite them.

PROBLEM ENCOUNTERED	SOLUTION IMPLEMENTED
One of the central dancers injured their legs 3 weeks before the performance.	We were able to fix this issue with the 6 extra dancers from Nrityalaya. Prisha spent extra time dedicated to teaching her the choreography and timing so she would be able to fulfill the role of the injured dancer.
In the first two weeks of ticket sales opening, we noticed that not as many people were purchasing tickets as we initially hoped.	To fix this, Saanvi advertised to various Tamil schools around the Greater Seattle Area and to the Nrityalaya School of Dance to broaden audience and propel ticket sales.
During the second month of rehearsals, we noticed that the dancers were having a hard time synchronizing in group dances.	To fix this, Prisha organized activities like movies, dinners, shopping, and sleepovers to strengthen bonds among dancers. Dancers brought the choreography to life by working together and dancing as one.

VI. CLOSING THE PROJECT

A. Evaluation of key metrics

To best assess our progress fulfilling our key metrics throughout the project, we established corresponding targets to which we could compare our outcomes. The outcomes and targets along with their relative metrics are displayed in the table below.

KEY METRIC	TARGET	OUTCOME	STATUS
Adherence to milestones (OTD)	0 delays	1 delay	⚠
Budget variance	$0	$0	✓
Number of ticket sales	450	501	✓
Number of attendees	450	812	✓
Customer satisfaction score	4	4.5	✓
Total funds raised	$10,000	$22,568	✓
Number of eye surgeries funded	187	423	✓

B. Lessons learned

Initiating

In the initiating phase, we were successful in our planned tasks. We received **financial assistance from WSSTA, selected specific and measurable objectives, and selected a fundraising medium in which we had expertise**. However, we had some **trouble coordinating with the international members of the team**. Keeping everyone updated proved to be difficult as many were on different schedules due to time zone variations.

Planning

In the planning and organizing, we were successful in establishing **attainable and timely objectives** for each key metric to accurately monitor progress, establishing an accompanying **timeline** to ensure we are on track to achieve objects, preparing a realistic **budget**, and establishing **backup plans** for mishaps. However, we did not plan for the **Microsoft Matching Gifts program**, which caused a **delay** in sending the funds.

Execution

During our execution, we were successful in using our **existing connections** (Nrithyalaya School of Dance) to source our artists, **advertising** through channels dedicated to South Asians, holding **2 programs** to maximize ticket sales, and using the **Microsoft Matching Gifts Program** to double our impact. However, we want to **broaden** and **diversify** our audience, and use more social media channels.

Monitoring

We were successful in establishing checkpoints and reviewing our budget. We could do better in **researching community programs beforehand** to stick to our schedule checkpoints.

Controlling

During our controlling phase, we were successful in **mitigating risks** we encountered throughout the project and **ensuring our team was on task**.

C. Recommendations for future projects

Running and managing the Parvaiyattam project has been an insightful experience, and one in which our team learned many lessons. For future projects, we recommend the following:

★ Research Existing Community Programs

Luckily, we found the Microsoft Matching opportunity after the event but before transferring the funds we raised. However, with this knowledge, our recommendation would be to research existing community programs to take advantage of them in order to maximize impact for future projects. We can also look for other community programs to identify areas where our goals align with theirs, and also establish partnerships with relevant organizations to reach a broader audience.

★ Expand Target Demographics

For the Parvaiyattam event, we prioritized marketing to a South Asian audience, since we believed they would be most likely to attend a Tamil cultural event and resonate with our cause. In future projects, we could diversify our target audience to increase our event audience, and maximize impact.

★ Utilize More Social Media Channels

We advertised the event with social channels, specifically Whatsapp and Facebook. However, for future projects, we should utilize more social media channels, like Instagram, to broaden our audience. By utilizing Instagram, we can target younger demographics. With a larger, more diverse audience, we would have greater ticket sales and therefore maximize our impact.

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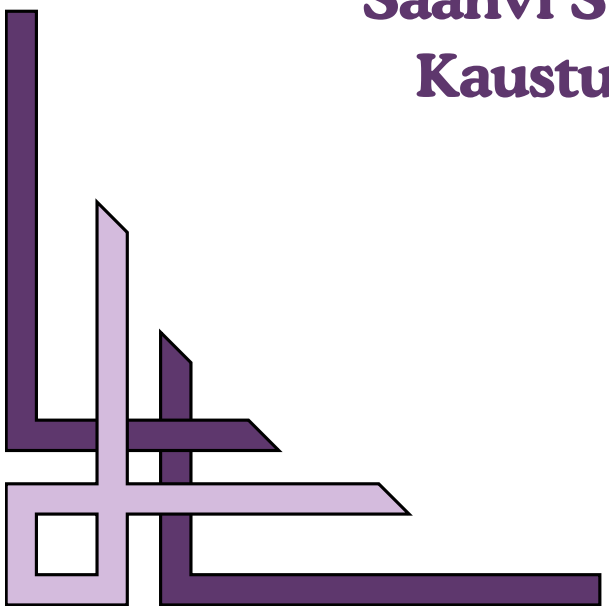
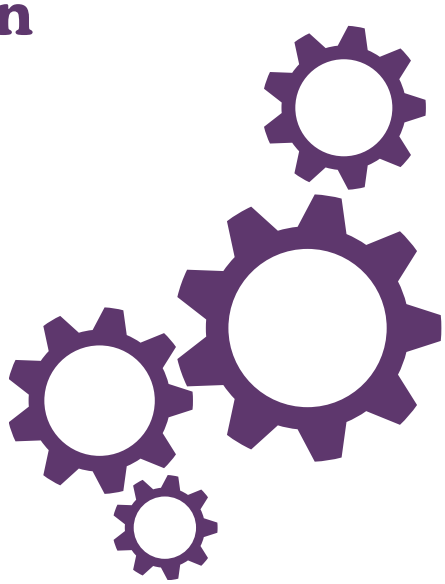
JANUARY 3, 2024

INTEGRATED MARKETING CAMPAIGN SERVICE

SWENext
Saanvi Subramanian
Kaustubh Grama





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I. EXECUTIVE SUMMARY

DESCRIPTION OF THE SERVICE

The Society of Women Engineers is a not-for-profit organization dedicated to supporting women interested in pursuing engineering. SWENext is their K-12 program to inspire students to pursue a career in STEM.

CAMPAIGN OBJECTIVES

INCREASE SWENEXT ENROLLMENT

- **25% increase in SWENext club member registrations**
- **20 new SWENext clubs in ≥10 new regions**

IMPROVE PARTICIPANT SATISFACTION

- **Partner with ≥15 collegiate chapters**
- **200+ mentor-mentee matches**
- **Mentorship average attendance rate: 80%**

ENHANCE SOCIAL MEDIA ENGAGEMENT

- **5,000 Instagram and TikTok followers**
- **1,500 campaign hashtag mentions/shares**
- **CTR: 3%**

TARGET MARKET

PRIMARY	SECONDARY
Girls in High School • **Grades 9-12** • **Ages 14-18**	Girls in Elementary and Middle Schools • Grades K-8 • Ages 6-13

CAMPAIGN ACTIVITIES AND SCHEDULE

#BeThatEngineer **3/1 - 3/30**

CURIE-OSITY KITS

- Engineering kits delivered to kids to build their own cool toys
- Partnership with Crunch Labs' Build Boxes
- Sent biweekly to SWENext clubs across the country

$10,528.00

BIG SIS MENTORSHIP

- Connects SWENext members with SWE collegiate members from various engineering fields

$0.00

WOMENGINEERING WEBINARS

- Matches SWENext members with collegiate mentors from nearby SWE chapters.
- Builds connections for guidance and support

$690.00

#BeThatEngineer

PROMOTION:

- TikTok and Instagram ads/posts
- Promote SWENext activities and opportunities

$450.00

CHALLENGES:

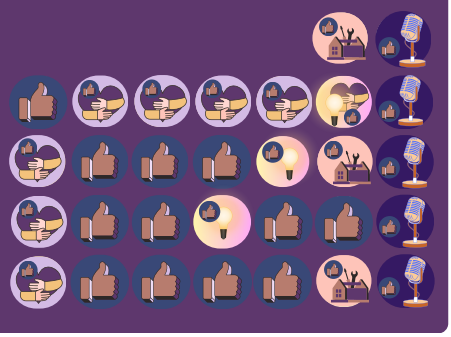
- Create an incentive for innovation
  - $1000 Scholarship for winners of 3 themed challenges

$9,000.00

CELEBRATE WOMEN'S HISTORY MONTH WITH SWENEXT!

MARCH 1-30TH

S	M	T	W	Th	F	S



KEY METRICS

- # of new member registrations
- # of new SWENext clubs
- # of new regions of clubs
- # of collegiate programs partnered with
- # of successful mentor-mentee matches
- Average attendance rate in career exploration efforts
- # of social media followers
- Campaign hashtag performance
- CTR for campaign content

CONCLUSION

SWENext is looking for **$20,668** to grow and enhance their K-12 program through the **#BeThatEngineer** campaign, to build a brighter, more diverse, and more inclusive future of STEM.

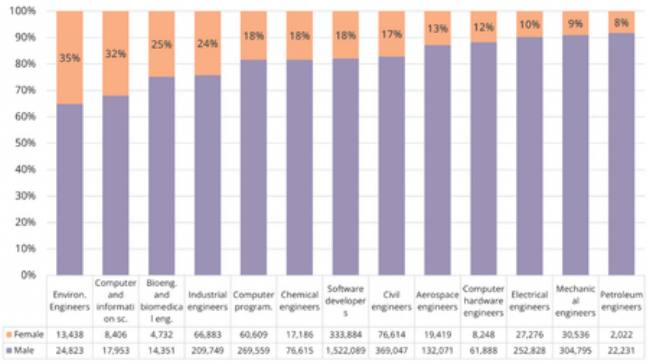
II. DESCRIPTION OF THE EVENT, PRODUCT, OR SERVICE

Our 30-day #BeThatEngineer marketing campaign is designed to grow and enhance SWENext, the Society of Women Engineers' K-12 program.

What is SWENext?

SWENext is a not-for-profit K-12 program sponsored by the Society of Women Engineers (**SWE**) collegiate organization designed to inspire more girls and those of underrepresented backgrounds to pursue STEM, with a focus on **engineering**, given the alarming gender disparities in the field. For more than seven decades, SWE has given women engineers a unique place and voice within the engineering industry, and SWENext aims to empower students to prepare for careers in engineering and technology and provide opportunities to learn and network with peers, role models, and engineering professionals. With locations across the globe, SWENext clubs can be started up by any club connected to the SWE organization, and can run through volunteer efforts. Any student can become "SWENexter" either through joining an existing club or starting their own, or registering for membership independently. To maximize their reach, SWENext is free to join and provides students access to programs, mentors, and resources designed to develop leadership skills and self-confidence to **#BeThatEngineer**.

FULL-TIME EMPLOYED ENGINEERS BY GENDER IN 2021



Research shows that girls often stop "seeing" themselves in the fields of math and science by the time they reach high school.

The long-term goal of SWE's K-12 Outreach programming is to open pathways for girls to become engineers and create gender equity in the field.

STRENGTHS

- Established reputation from SWE
- Global reach
- Offers mentorship and networking resources

WEAKNESSES

- Limited resources as a not-for-profit
- Volunteer-dependent

OPPORTUNITIES

- Growing demand for STEM education
- Collaborations and partnerships
- Opportunities to grow online engagement

THREATS

- Competition with other STEM initiatives
- Resource constraints with limited funding
- Retention and engagement (sustaining interest)

III. CAMPAIGN OBJECTIVES

INCREASE SWENEXT ENROLLMENT

We aim to significantly expand the reach of SWENext by boosting the number of K-12 members. Currently there are **5,000 SWENext members, 350 SWENext clubs in 200 different regions**. In this campaign, we seek to encourage more students to join SWENext clubs, helping them explore engineering and STEM fields, with specific goals to increase these numbers to **6,250 SWENext members, 370 SWENext clubs in 210 regions in total**.

Achieve a 25% increase in SWENext club member registrations within the 30-day campaign period

Establish 20 SWENext clubs in at least 10 new regions or countries

IMPROVE PARTICIPANT SATISFACTION

This objective focuses on enhancing the overall experience of SWENext members by providing more supportive interactions by creating mentorship opportunities through collegiate chapters. Currently, SWENext is officially associated with **50 collegiate chapters**, and we aim to increase this to **65 collegiate chapters**, create direct mentorship for **200 SWENext members**, and mentorship check-ins with an **average attendance rate of 80%**

Partner with 15+ collegiate chapters for mentorship

Facilitate mentor-mentee matches for 200+ students within the campaign period

Mentorship average attendance rate of 80%

ENHANCE SOCIAL MEDIA ENGAGEMENT

To increase the program's online presence and audience engagement, we aim to utilize social media platforms to boost SWENext. Currently, their Instagram account has **1.7K followers** and there is no SWENext TikTok. With our targeted posts and advertising programs, we aim to achieve **5K followers** on both platforms, **1,500 mentions/shares with the campaign hashtag** (#BeThatEngineer), and a **Click-Through Rate of 3%**.

Achieve 5,000 followers on Instagram and TikTok

1,500 campaign # mentions/shares

CTR: 3%

IV. CAMPAIGN TARGET MARKET

DEMOGRAPHICS

- High school girls
  - 9-12th grade
  - 14-18 years old
  - All races and income levels

GEOGRAPHICS

- Rural to urban high schools across the US
  - Areas near collegiate chapters

PSYCHOGRAPHICS

- Students interested in problem-solving, design
  - May or may not be exposed to engineering

BEHAVIORS

- Use social media
- Participate in STEM clubs
  - Exploring various career paths
  - Excel in math and/or science
  - Have friends interested in STEM

PRIMARY

PROFILE

HI, I'M JESSICA! I'm a 10th grade student at Vista Green High School in Bellevue, Washington. I've always like creating and designing things, but I've never been had the chance to explore the field of engineering yet.

DEMOGRAPHICS

- Elementary and middle school girls
  - K-8th grade
  - 6-13 year old

GEOGRAPHICS

- Near elementary and middle schools
- Areas dense in child populations

PSYCHOGRAPHICS

- Students exploring their interests
- Involved parents
- Engineer relatives

BEHAVIORS

- Participate in after school clubs
- Enjoy hands-on activities
  - Arts and Crafts

PROFILE

HI, I'M CLAIRE! I'm 8 years old in Juanita Elementary School in Bothell, WA! I like science!

SECONDARY

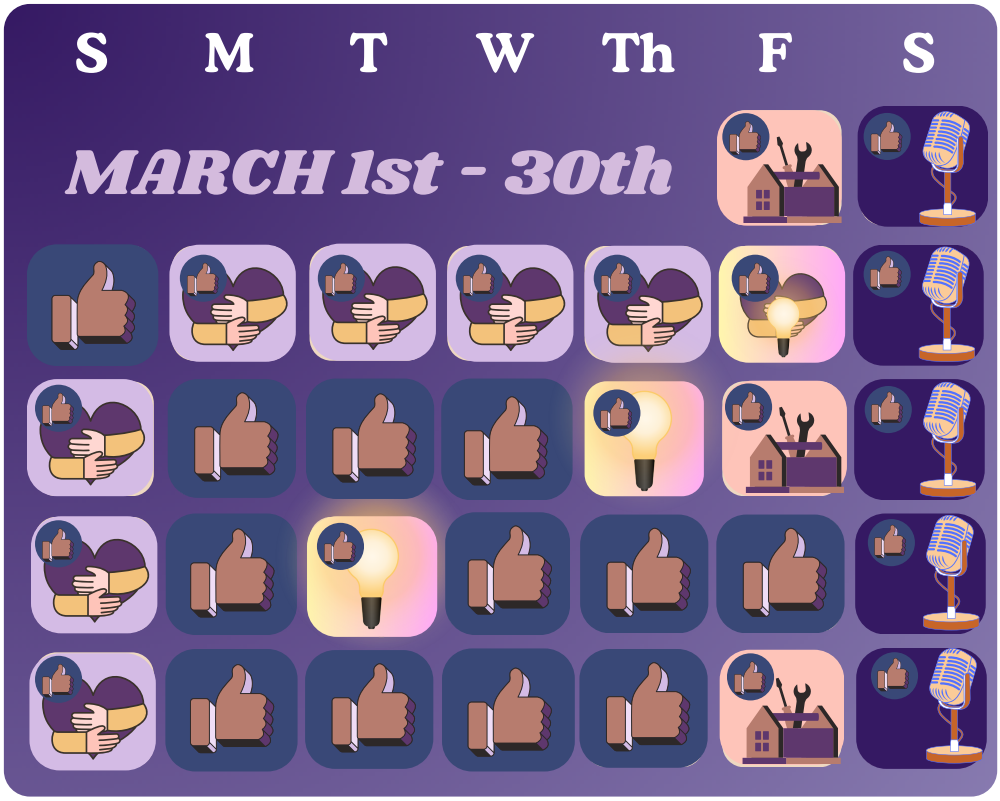
V. CAMPAIGN ACTIVITIES AND SCHEDULE

CELEBRATE WOMEN'S HISTORY MONTH WITH SWENEXT!

The **#BeThatEngineer** campaign comprises of the following four activities throughout March, *Women's History Month.*

KEY

- **CURIE-OSITY KITS**
- **BIG SIS MENTORSHIP**
- **WOMENgineering WEBINARS**
- **#BeThatEngineer PROMOTION**
- **#BeThatEngineer CHALLENGES**



CURIE-OSITY Kits

Kicking off the #BeThatEngineer campaign, SWENext will roll out CURIE-OSITY kits to SWENext clubs nationwide **every other Friday**. Paying homage to famous scientist Marie Curie, CURIE-OSITY kits are designed to challenge girls to think like an engineer, empowering them to achieve whatever they set their minds to. From building robots to rockets, CURIE-OSITY kits aim to fuel creativity and encourage hands-on exploration of engineering concepts, with different themes of toys each delivery.

Created by NASA engineer and YouTube personality, Mark Rober, Crunch Labs' boxes are designed for kids 8-12+ so they can build toys successfully and independently, and learn engineering concepts in the best possible way – by playing! **CURIE-OSITY kits are created in partnership with Crunch Labs and are special-edition versions of their Build Boxes that are made to appeal to girls.** It will take 2-5 days to ship the kits, and SWENext will distribute them via direct mail to SWENext clubs across the nation free of cost for the members using USPS Ground Advantage shipping.

The themes for the three kits sent out during the campaign are **robotics, sustainable energy, and rockets**! Each CURIE-OSITY kit consists of a DIY toy along with an instruction manual, basic building tools, and a QR code which can be scanned to unlock a special 15-30 minute video from Mark Rober which explains the toy through tutorials and goes in-depth with the mechanics that allow the toy to function. For example, the CURIE-OSITY kit focused on sustainable energy will direct users to a video about sustainable energy, including how wind turbines work, what it means for energy to be sustainable, and applications in the real world.

BI-WEEKLY DISTRIBUTIONS

MARCH 1ST	MARCH 15TH	MARCH 29TH
Build a robot!	**Create a wind turbine!**	**Launch a rocket!**

WHAT'S INCLUDED?

- A new build-it-yourself toy every 2 weeks
- Unlock a fun and educational 15-30 minute video from Mark Rober with every box that walks through each toy and the engineering concepts behind them

OBJECTIVE: IMPROVE PARTICIPANT SATISFACTION, INCREASE SWENEXT ENROLLMENT

Big Sis Mentorship

During this campaign, we will **launch** our new Big Sis Mentorship program, where we match SWENext members with a volunteer **collegiate mentor**, from a neighboring SWE collegiate chapter.

The Big Sis Mentorship program aims to create a bridge between high school girls in SWENext clubs and collegiate mentors who are part of a collegiate Society of Women Engineers (SWE) organization. By connecting young girls with experienced professionals who have walked a similar path, we hope to offer valuable guidance, support, and inspiration.

One of SWE's core values is mutual support, meaning that the organization fosters mentoring and the development of professional and personal networks. SWE firmly believes that mentorship is a critical component in the success of women in engineering. SWE boasts several mentorship programs, including GradSWE, a support community connecting graduate women in engineering at different levels, SWEsters, which connects undergraduate underclassmen with upperclassmen and graduate students usually in the same department, and the SWE Industry Mentor Network designed to connect SWE collegiate chapter members with industry professionals to secure internships and job opportunities. However, the SWENext Big Sis Mentorship program will be the first official SWE mentorship program directly targeted at helping connect pre-college SWENext high school club members with SWE mentors from a college close to them.

Before the #BeThatEngineer campaign starts, SWE collegiate chapters will be contacted to find 20-30 volunteering members from each chapter willing to participate in this outreach program. The first week of the Big Sis Mentorship program will be dedicated to matching students to their preferred mentors. We will send out a secure survey to SWENext members to get to know them and their backgrounds and their interests in engineering. Based on each survey, members will be matched with a mentor with a similar background as them and studying the same department of engineering that the member selected they were interested in. If a SWENext member is unsure of what type of engineering they are looking to pursue, they will also have the opportunity to network with the mentors at the collaborating university to see who they would like to be paired with. There will be scheduled weekly check-ins each Sunday where mentors can arrange an activity with their students and answer any questions they may have. Students and mentors may meet additionally if they like. We will track the attendance rate of these check-ins as a metric of the success of the program, and a successful mentor-mentee match will be deemed by whether each match has a 100% attendance rate in their check-in sessions.

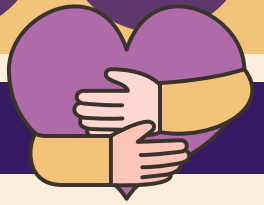
SWE COLLEGIATE CHAPTERS INCLUDE:

BIG SIS MATCHING PROCESS

MARCH 4TH - MARCH 8TH

BIG SIS CHECK-INS

- MARCH 10TH
- MARCH 17TH
- MARCH 24TH



OBJECTIVE: IMPROVE PARTICIPANT SATISFACTION

WOMENgineering Webinars

Every Saturday, SWENext will host a live, interactive webinar through Zoom, featuring female industry professionals from a variety of engineering disciplines, allowing the opportunity for girls to connect with real-life engineers and ask any questions they may have. Each webinar will last be 90 minutes (6:00 - 7:30 PST) with an hour-long session, and 30 minutes for a QnA. These webinars will be promoted through the SWENext mailing list, which consists of all SWENext members, and through social media ads **(#BeThatEngineer Promotion)**. WOMENgineering Webinars are free to join for any pre-college students, but there is a limit of 1000 people who can join each webinar.

WOMENgineering Webinars are designed to help build self-confidence and resilience among pre-college students interested in engineering and technology degrees and provides multiple opportunities to network with peers, mentors, role models, and industry professionals. With so many different secondary education and career paths to choose from, it can be overwhelming to determine what path is best for you. Each webinar will center around a distinct theme centered around the journey into engineering, and will highlight a different domain of engineering to broaden participants' understanding about the plethora of pathways in engineering.

Some of the featured speakers from the panel of WOMENgineers include:

Dr. Jennifer Doudna, Bioengineer and nobel Prize-winning biochemist, acclaimed for her groundbreaking contributions to CRISPR-Cas9 gene editing technology.

Dr. Ayanna Howard - Electrical engineer and roboticist, known for her contributions to robotics and artificial intelligence.

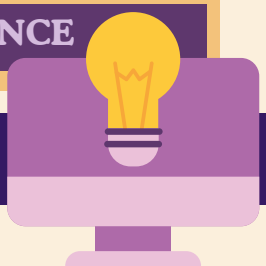
Dr. Linda Katehi - Mechanical engineer and former Chancellor of the University of California, Davis, recognized for her efforts in empowering women in STEM.

Dr. Jocelyn Bell Burnell - Astrophysicist and aerospace engineer, credited with the discovery of pulsars.

Dr. Fei-Fei Li - Computer scientist and AI researcher, known for her work in computer vision and directing the Stanford Institute for Human-Centered AI

WOMENGINEERING WEBINAR SCHEDULE

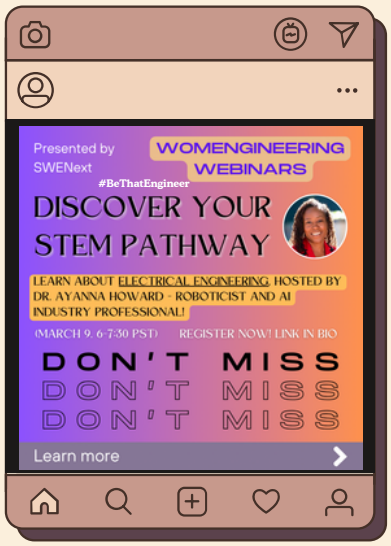
MARCH 2nd: WHY ENGINEERING MATTERS	CIVIL ENGINEERING
MARCH 9th: DISCOVER YOUR STEM PATHWAY	ELECTRICAL ENGINEERING
MARCH 16th: ART, SCIENCE, & ENGINEERING	MECHANICAL ENGINEERING
MARCH 23rd: DIVERSITY AND INCLUSION	AEROSPACE ENGINEERING
MARCH 30th: CODING THE FUTURE	COMPUTER SCIENCE



OBJECTIVE: IMPROVE PARTICIPANT SATISFACTION

#BeThatEngineer Promotion

We plan to roll out social media ads on **Instagram** and **TikTok** to promote SWENext. These platforms boast diverse user bases with a significant presence of young women, which makes them ideal for reaching our target audience. Additionally, Instagram is great for sharing stories, pictures, and longer videos that people can share easily. We also plan to use TikTok, for its potential for posts to go viral and its dedicated STEM page which promotes STEM-related content.



Sample Instagram ad promoting WOMENgineering Webinar 2

These social media ads will include a variety of content to promote SWENext and the different activities they host, including our **WOMENgineering Webinars**. On the left is a sample of one of these Instagram ads that would be used to promote the webinars. To ensure we reach a large audience, we will be promoting posts like these through Instagram and Tiktok's advertisement features. We will put out **15 Instagram ads** and **15 TikTok ads.**

Additional ad content material includes inspirational stories from famous women scientists and engineers, **#BeThatEngineer Challenges**, resources and special opportunities available for young women through the SWENext program to highlight its supportive community promoting diversity in STEM.

SWENext's advertisements will run throughout the entire #BeThatEngineer campaign, from March 1st to March 30th; continuous promotion.

OBJECTIVE: INCREASE SWENEXT ENROLLMENT

#BeThatEngineer Challenges

As part of our #BeThatEngineer social media campaign, we're excited to introduce three holiday-themed challenges for the month of March! Participants will submit their entries to SWENext through a Google form with links to their posts. As an incentive for challenge participants, the top 10 submissions selected by SWENext committee will be invited to present their creations at a virtual SWENext symposium comprising collegiate mentors, industry professionals, and the SWENext national leadership committee! There, final judging will be held, and the top 3 contenders of each challenge will win a **$1,000 ScholarsInStem scholarship** - provided by SWENext.

MARCH 8TH

INTERNATIONAL WOMEN'S DAY

InventHER Spotlights

In honor of International Women's Day, we are excited to launch the ***#InventHER*** Spotlights campaign, showcasing the remarkable accomplishments of female engineers and SWENexters. Participants will share a 5-minute video explanation of their chosen invention by a woman! Submissions will be evaluated by SWENext for symposium qualification.

MARCH 14TH

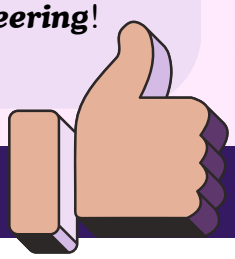
PI DAY

With the hashtag ***#PIoneering*** challenge, we're spicing up math with creativity. Participants will be challenged to create a 5 minute video entry explaining the engineering principles behind making a pie, like heat transfer and structural integrity for example. Submissions will be evaluated by SWENext for symposium qualification.

MARCH 19TH

SPRING EQUINOX

With the arrival of spring, we're launching a challenge that centers around environmental engineering and sustainability. Participants will be encouraged to share their projects illustrating how engineering can contribute to a greener and more sustainable world with the hashtag ***#GREENgineering***! Submissions will be evaluated by SWENext for symposium qualification.



OBJECTIVE: ENHANCE SOCIAL MEDIA ENGAGEMENT

VI. BUDGET

Campaign Activity	Item	Cost Per Item	#	Total Cost
CURIE-osity Kits	Kit	$21.57	400	$8,628.00
	USPS Ground Advantage™ Shipping	$4.75	400	$1,900.00
Big Sis Mentorship	No Direct Cost; Volunteer-based			$0.00
WOMENgineering Webinars	Zoom Webinar License	$690.00	1	$690.00
#BeThatEngineer Promotion	Instagram Post Promotion	$15.00	15	$225.00
	TikTok Post Promotion	$15.00	15	$225.00
#BeThatEngineer Challenges	ScholarsInStem Scholarship	$1,000.00	9	$9,000.00
GRAND TOTAL				$20,668

RETURN ON INVESTMENT

The true value of the $20,668 investment lies in empowering future female engineers, expanding global presence, and fostering mentorship and support networks. CURIE-osity Kits engage SWENexters with the tangible aspect of engineering, inviting creativity and innovation. Big Sis Mentorship exposes SWENexters to the various engineering disciplines and the plethora of opportunities available to them. Social media promotion enhances outreach, and themed challenges offer incentives for innovation. In the long term, this investment will drive positive change and increase the number of women pursuing engineering, leading to a brighter, more inclusive future in STEM.

VII. KEY METRICS

INCREASE SWENEXT ENROLLMENT

For this objective, the key metrics for success are the **number of new SWENext member registrations**, **number of new clubs established**, and **number of new regions** where clubs are formed. For the #BeThatEngineer campaign, the goal is to achieve a 25% increase in member registrations (1,250 new members), and to establish 20-30 new SWENext clubs in at least 10 new regions, broadening global engagement and encouraging diverse participation.

UNDER-PERFORMING	MEETS EXPECTATIONS	EXCEEDS EXPECTATIONS
15-25% increase in members	**1,250 new club members**	**30+% increase in members**
10-20 new SWENext clubs	**20-30 new SWENext clubs**	**30+ new SWENext clubs**
0-10 new club regions	**10-12 new club regions**	**12+ new club regions**

IMPROVE PARTICIPANT SATISFACTION

To improve SWENext participant satisfaction, the following key metrics have been identified: **number of collegiate programs** partnered with, the **number of successful mentor-mentee matches**, and **average attendance rate** (tracked through survey form) in mentorship check-ins. The goal is to partner with 15 collegiate programs and facilitate 200 successful matches, and an average attendance rate of 80%, from those who have registered.

UNDER-PERFORMING	MEETS EXPECTATIONS	EXCEEDS EXPECTATIONS
Partner with ≤10 college programs	**Partner with ≥15 college programs**	**Partner with ≥20 college programs**
150 mentor-mentee matches	**200 mentor-mentee matches**	**250 mentor-mentee matches**
70% average attendance rate	**80% average attendance rate**	**90% average attendance rate**

ENHANCE SOCIAL MEDIA ENGAGEMENT

The key metrics for this objective are **number of social media followers**, **hashtag performance** to evaluate the effectiveness of #BeThatEngineer, and **click-through rate** for campaign content to assess how effectively the content is driving engagement. Our targets are 5,000 followers on Instagram and TikTok, 1,500 mentions and shares with the campaign hashtag, and a CTR of 3%, slightly above the industry average of 2.7%.

UNDER-PERFORMING	MEETS EXPECTATIONS	EXCEEDS EXPECTATIONS
<4,000 Instagram/ TikTok followers	**5,000 Instagram/ TikTok followers**	**6,000+ Instagram/ TikTok followers**
<1,000 mentions/ shares with #	**1,500 mentions/ shares with #**	**>1,500 mentions/ shares with #**
CTR <2.5%	**CTR ~3%**	**CTR >3.5%**

VIII. BIBLIOGRAPHY

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